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A review on recent development for the design and packaging of hybrid photovoltaic/thermal (PV/T) solar systems

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ABSTRACT

The design of high performance photovoltaic/thermal solar system involves many parameters related to thermal, economic and reliability issues. The current paper provides a comprehensive review for the recent studies that have discussed the design and cooling considerations to build a high performance hybrid Photovoltaic/Thermal (PV/T) system. The review highlights the importance of the nanofluids and nano phase change materials (nanoPCMs) as an alternative for the traditional ways of cooling and storage. In addition, the application of the nanofluids in optical filtration has been covered. The main conclusions that have been reached at the end of the review show that cooling from both sides of the PV, with involvement of the optical filtration using nanofluid; will result in high improvements in the performance of the PV/T system. In addition, change of the channels cross section to honeycomb and induction of nanoPCM for storage and cooling purposes can provide better distribution of the thermal load.

1. Introduction

Hybrid Photovoltaic/Thermal (PV/T) collector which combines between the collection of thermal energy and generation of electrical energy is a promising solution to achieve a better exploitation for the solar energy. PV/T collectors can produce more energy per unit surface area in comparison with PV modules and solar thermal collectors [1]. Only around 15–20% of the solar radiation received by the PV panels is converted to electrical output and the rest of the solar energy ends up heating the PV cells [2], which results in several problems. Furthermore, the operating temperature that can be reached by a PV panel and as a result of the generated electric power is strongly affected by the climate of the site [3]. One of the highly affecting parameters on the operating temperature and the power drop from a PV system is the climatic conditions. In Germany about 50% of solar radiation is above 600W/m^2, while in Sudan around 80% of solar radiation is above the same value [4,5]. Moreover, degradation in the electrical efficiency occurs for a crystalline silicon PV operating at 25 °C with a rate of 0.45–0.6% for each degree increase in cell temperature above its nominal operating temperature. The exact rate depends on the PV material type [6–9] and the climatic conditions as mentioned before, which means very high degradation of electrical efficiency in countries which suffer harsh weather and hot ambient temperatures especially in summer season. The most critical period for PV panel efficiency degradation is in regimes of highest solar irradiation levels and lowest wind air velocities [10]. The other problem that can result from the high cell temperature is the capability of reaching the full damage as a result of overheating [11]. As a result, cooling of the PV panels is a necessary task that should be taken seriously into consideration. Cooling of PV can be used for heating of a fluid at the same time so we can improve the electrical efficiency and at the same time cultivate some thermal energy. This combination gave rise of hybrid system. Many types of hybrid collectors are available, where the PV is the common part. These types like the PV/T flat plate collector, PV/T concentrator collector, and PV/T heat pipe collector. The most popular collector type in the recent years is the hybrid PV/T flat plate solar collector. The PV/T collector is composed of the PV panels, the absorbing surface which is mainly flat plate, the flow channels, working (cooling) fluid, and the storage medium if required. Each one of these components may affect the performance of the PV/T.

This paper proposes a comprehensive study on each of the previous parameters showing different approaches for the design, PV materials, PV-absorber attaching methods, cooling methodologies, cooling fluids, and flow channels. Fig. 1 shows the skeleton of the proposed review,

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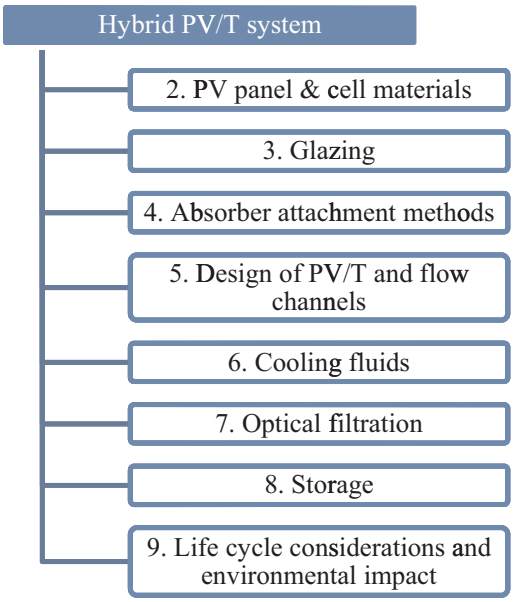


Fig. 1. Flow chart for the skeleton of the paper.

which will be followed by a case study in Section 10 to assure the importance of the study. To the best of the authors' knowledge, this is the first paper that presents all the design consideration for hybrid PV/T to be used as a guideline for researchers and engineers in the field.

2. PV panel and cell materials

Recently, it can be said that the most commonly used photovoltaic technologies are those based on silicon like mono-or poly-crystalline materials [12–14]. The general characteristic for the considered PV panels (mono and poly) are shown in Table 1. No significant difference between mono and poly- crystalline PV technology was noticed in their corresponding thermal profiles.

The typical crystalline silicon cell can absorb incident light in 300–1100 nm range due to the band gap of thin-film silicon [15]. It can be considered that −40–85 °C is the suitable operating temperature range for a PV panel, but this range is not fixed as in harsh and hot weathers, the temperature may increase above this limit [16]. Regarding the industrial production, the manufacturing of mono-crystalline silicon-based PV cells shares around 80% of the total cells production [17], and theoretically their efficiency can be up to 28% [18], but actually it doesn't exceed 17% [19]. Because of thin silicon layer and the importance of maintaining the cells in good conditions to keep its performance at high levels. In this context, good operating conditions include structural integrity free of stresses, cell temperature as close as possible to optimum values, avoiding hot spots, which can significantly drop the efficiency and proper glazing surface-finish to reduce dust collection. There was a need to protect these PV cells by

Table 1
General characteristic for some considered PV panels (mono and poly) (reprinted from [10], with permission from Elsevier).

Basic technical data for PV[a] panels	Mono	Poly
Model	SL−50AA36	SL-50P
Number of cells in the module	36	36
Maximal power output	50 W ± 3%	50 W ± 3%
Maximal power voltage/Open circuit voltage	17.2 V/22.9 V	18 V/22.5 V
Maximal power circuit current/Short circuit current	2.91 A/3.18 A	2.78 A/3.02 A
Panel area	0.357 m^2	0.39 m^2

[a] PV panel characteristics at irradiance of 1000 W/m^2 and 25 °C.

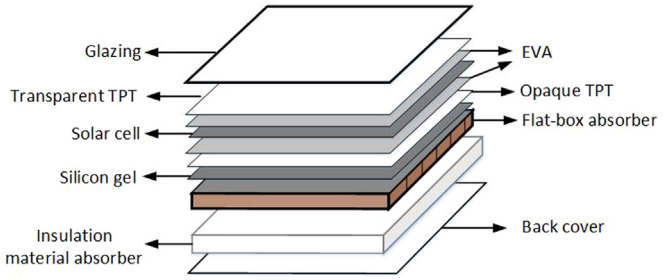


Fig. 2. Layers of Hybrid PV/T system (redrawn from [20]).

other layers that should has high thermal conductivity, low electrical conductivity, and high toughness. Fig. 2 shows a schematic diagram for the different layers that are surrounding the solar PV cells and the attachment with the absorbing channels forming the hybrid PV/T system. The main two layers are the EVA and TPT layers. The EVA layer is a thermoplastic containing ethylene vinyl acetate and is used to encapsulate the photovoltaic cells. TPT is the abbreviation of Tedlar Polyester Tedlar and is a composite-layer film used as covering foil on the backs of solar modules. Silicon gel is an adhesive material that is used to connect between the PV module and the absorbing channel, and it should have the capability to withstand fluctuation of the system temperatures between high and low temperature as well as expansions resulted from high temperatures.

PV has been modified with time and the most commonly used material for production was the monocrystalline silicon. Monocrystalline cells showed better efficiency than the multicrystalline ones. Due to the low efficiency of PVs, protection of the cells from any damage sources is very important to maintain its performance. Usage of EVA and TPT has protected the cells from any environmental effect, but the problem of the high temperature and the possibility to have hot spot still in a need for more research work.

3. Glazing

Glazing is a very important constituent of any PV-based system that helps in protecting the panels from several environmental effects. In contrast, glazing may prevent part of the solar radiation to arrive at the surface of the PV cells and the amount of prevention depends on the transmissivity of the glazing used. Many glazing layers can be used in front of the PV panel especially if the cooling of the PV will be from the top side. Under this condition, the glazing layers behave as a channel for the cooling fluid. Using many layers with cooling is beneficial as it serves for cooling of the PV cells while at the same time it obstructs the solar radiation from reaching the PV. Optimization of the number of glazing layers was required. It can be said that the best energy performance can be attained using single glazing, while the best exergy can be attained without any glazing [21]. Glazing is normally important to maximize the amount of thermal energy output. In contrast, glazing is not preferred when high electrical energy generation is required. So, it can be said that adding a cover on the top is important to prevent thermal losses but this cover should also be highly transparent with lowest possible refractions of solar incident radiation [22]. Riffat and Cuce [20] provided a comprehensive study on PV and PV/T solar collectors. They reported that using glazing in the PV/T helps in maximizing the thermal and the overall energy output. In contrast using bare PV/T without glazing helps in enhancing the electrical efficiency as compared to the glazed one due to not scattering the irradiation.

Then, it can be claimed that increasing number of glazing covers, higher thermal energies should be attained. However, Zondag et al. [23] demonstrated that using more than 3 layers of glass covers with PV/T systems is not recommended as that will cause low transmittance to the solar irradiation, although it will enhance the thermal efficiency, hence 3- glazing covers is the limit. The double glass cover is considered

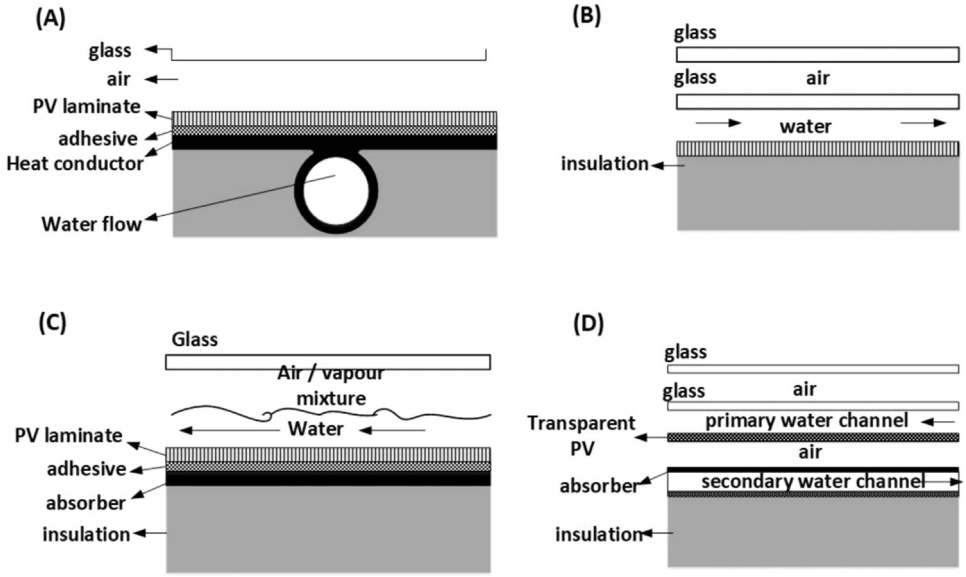


Fig. 3. Various collector concepts: (A) sheet-and-tube PV/T, (B) channel PV/T, (C) free flow PV/T, (D) two-absorber PV/T (insulated type) (redrawn from [23]).

better than the single glass cover in terms of the overall energy performance [24].

Glazing has been considered as extra protection for the PV cells from environmental effects with some drawbacks related to the absorption of solar radiation. It allows for more thermal energy collected using different cooling fluids in hybrid PV/T systems. Three glazing covers is the limit beyond which the PV/T systems are not expected to be efficient.

4. Absorber attachment methods

Construction of PV/T is mainly done by attaching a commercially available PV layer to a thermal absorber. The absorber plate and its attaching criteria are very important in affecting the thermal efficiency of the collector [20]. This attachment can be made using integration methods which are either mechanical or chemical [25].

There must be a gap filler in between the PV panel and the thermal absorber and it should compensate for thermal expansion of different parts in the PV/T. This gap filler should facilitate for high solar absorption and low heat resistance, so increasing the heat removal factor [25]. Resistivity for heat transfer from PV panel may result in high temperatures [22]. Jee et al. [22] mentioned that resistivity could be avoided using highly conductive materials between the PV and the absorber below. Moreover, direct attachment between PV and absorber could enhance the thermal efficiency by more than 50%, which was not an easy matter to be achieved due to air gaps problems.

Air gaps and air bubbles shouldn't be allowed in-between the two layers in order not to have extremely large heat removal resistance [25]. Jinshun et al. [25] did a review on the thermal absorbers and their corresponding integration methods. Micro-channel heat pipe array/heat mat, extruded heat exchanger, roll-bond heat exchanger and cotton wick structure, were promising as compared to the traditional absorbers like sheet and tube structure, rectangular tunnel with/without fins/grooves and flat-plate tubes. Ethylene vinyl acetate (EVA) lamination method was found to be the best suitable for the integration purposes of the flat plate absorber to the PV panels when compared with other conventional methods, such as direct contact, thermal adhesive and mechanical fixing. Extruded metal heat exchanger is the most suitable for solar collectors giving simple manufacturing and low cost [25].

Attachment of the absorber channels to the PV plays a very important role in enhancing the efficiency of the PV/T systems. Different methods exist for this purpose. Thermal expansion of different parts should be taken into account in selecting the attachment method. In addition, the air gaps and bubbles shouldn't be allowed to exist between the PV and the absorber as it increases the resistance for heat transfer. EVA was found to be the best suitable for the integration purposes of flat plate absorber to the PV panels.

5. Design of PV/T and flow channels

Many studies showed that the efficiency of the system increased with the increase in collector length, mass flow rate and cell density while, it decreased with an increase in duct depth [20]. Consequently, flow channels shape, size, location, and arrangements were found to be highly effective in increasing or decreasing the possible heat transfer or cooling of the PV/T, which was reflected on the performance of the collector. Many channel shapes were studied depending on the flowing fluid. Starting from simple channels or ducts of air and the possibility to enhance heat transfer using fins and use of water in cooling and the capability of using pipes with different sizes and spacing instead of channels. Studies also discussed how these pipes can be arranged to attain the highest possible cooling and as a result thermal efficiency; direct pipes, serpentine, spiral … etc. Improvements transferred to changing the shapes of the pipes from circular cross-section to half circular or triangular were also discussed. The effect of the location of these channels above and below the PV module and the effect of the contact length of the channels with the PV module on the heat transfer were studied as well.

An old study proposed by Zondag et al. [23] studied nine different PV/T designs considering the evaluation of the energy-generation efficiency. These nine different designs were classified in four groups; sheet and tube, channel, free flow, and two-absorber PV/T collectors, Fig. 3. Each group consists of different configurations considering the glazing, insulation, and the channel location relative to the PV module. Using the insulated type of the two-absorber PV/T collector, the highest average thermal efficiency in a year was reached with significantly low electrical efficiency relative to the other designs. In contrast, using the uncovered sheet and tube PV/T collector, the highest average electrical efficiency in a year was reached with the lowest thermal efficiency due to high heat losses. Generally, Zondag et al. [23] have stated that the channel-above-PV design, the two-cover sheet-and-tube design and the insulated two-absorber provided the highest thermal annual profit with lower electrical output due to glazing. within conclusion, they stated

that from efficiency point of view; channel-below-transparent-PV seemed to be the best option.

Fudholi et al. [26] investigated different configurations of water cooled PV/T under different mass flow rates and solar irradiation levels. They proposed three different new designs for the absorber channels; web flow absorber, direct flow absorber, and spiral flow absorber. As mass flow rate increased, the PV temperature decreased and the electrical efficiency increased. In addition, increasing the solar radiation level increased the PV temperature and the generated electrical efficiency. They notified that the behaviors of the direct flow and the spiral flow absorbers were comparable; higher thermal efficiencies can be attained using the direct flow absorber while higher electrical efficiencies can be obtained using spiral flow absorber. The spiral flow absorber presented the highest overall performance at solar incident radiation of 800 W/m2, with overall, thermal, and electrical efficiencies of 68.4%, 54.6%, and 13.8% respectively. Hussain et al. [27] studied the efficiency of using aluminum honeycomb design heat exchanger beneath the PV in enhancing the thermal efficiency of hybrid PV/T collector cooled by air. They examined the enhancement under different flow rates. Using honeycomb channels, the thermal efficiency has enhanced from 27% to 87% at air mass flow rate of 0.11 kg/s. The enhancement in the electrical efficiency was almost negligible (0.1%) which may be attributed to using air in cooling, which had low thermophysical properties. This innovation was easy for manufacturing process and compact, which enables it to be implemented easily for building integration (Figs. 4-6).

Yusof et al. [28] investigated three different configurations for an air cooled PV/T solar collector; the V-groove, honeycomb, and stainless steel wool as shown in Fig. 7. The honeycomb design provided the best performance among the three. It provided the maximum outlet fluid temperature as a result of high heat transfer. Honeycomb design provided the highest thermal efficiency, which increased as mass flow rate increased till a certain limit and after that it decreased slightly and remained almost constant. The stainless-steel wool heat exchanger had a very close behavior to that of honeycomb heat exchanger. In addition, electrical efficiency was dependent in direct proportion on the mass flow rate. Using the honeycomb design at incident radiation of 828 W/m^2 and mass flow rate of 0.11 kg/s, the thermal and electrical efficiencies were found to be 87% and 7.13% respectively.

Regarding changing the design of the PV/T system, different studies that based on changing the design of the system to enhance the cooling process are discussed next.

Rahimi et al. [29] carried out an experimental study to test the effect of using a cooling device that was based on a conic wind-collecting tunnel with the PV system. For the cooling device design, the authors aimed to provide a cooling technique to the PV system by spraying the coolant on the cells and at the same time generate electricity from the rotation of the turbine because of wind induction through the tunnel as depicted in Fig. 8. The total output has increased by about 36% because of the electric output from both the PV and the turbine.

Touafek et al. [30] presented an experimental as well as numerical study to study the enhancement in overall power output of a hybrid PV/T system as a result of using an enclosure-shaped absorber as in Fig. 9. They studied the temperature distribution through the different layers

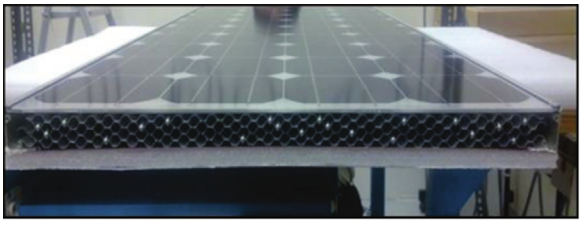


Fig. 4. Photo of honeycomb installed at the back of the PV module (reprinted from [27], with permission from AIP Publishing LLC).

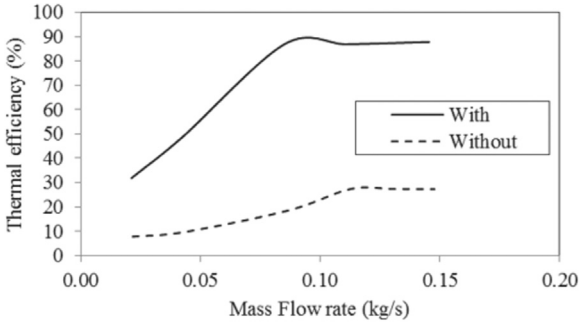


Fig. 5. Thermal efficiency, of both collectors (reprinted from [27], with permission from AIP Publishing LLC).

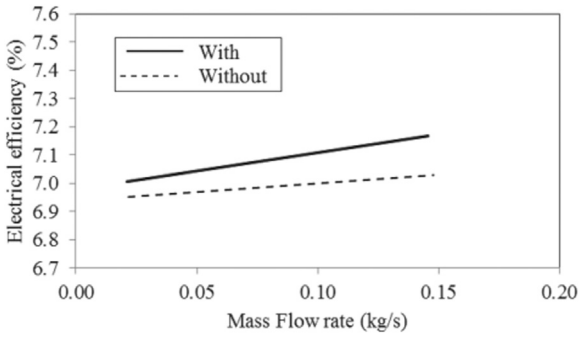


Fig. 6. Electrical efficiency, of both collectors (reprinted from [27], with permission from AIP Publishing LLC).

of the system and the overall efficiency was calculated to be around 80%. The new absorber design helped in improving the thermal performance of the hybrid system if compared to the traditional hybrid PV/T system with single absorber layer. The group did not cover the economic feasibility of the new design.

Mojumder et al. [31] conducted out an experimental study on the effect of attaching thin rectangular fins for heat dissipation on the performance of an air cooled hybrid PV/T system shown in Fig. 10. Under different fin numbers (0 − 4), air mass flow rates (0.02 kg/s–0.14 kg/s), and solar radiation intensities (200 W/m2–700 W/m2), the temperature distribution along different layers were examined. The authors recorded that using four fins, with 0.14 kg/s of air and solar radiation of 700 W/m^2, the system could work under the maximum thermal and electrical efficiencies (56.19% and 13.75% respectively). Generally, attaching fins has enhanced the overall performance of the PV/T system as compared to the case without fins. Increasing the fin number, air mass flow rate, and solar radiation intensity helped in enhancing the overall performance of the system as shown in Figs. 11 and 12.

Xu et al. [32] used a super-thin absorber with PV panels to form a hybrid PV/T system with enhanced output relative to the conventional PV system. Experimentally, the group tested 15 traditional PV panels against 15 from the new provided PV/T systems. Usage of the new systems enhanced the electrical output by around 5% and a thermal efficiency about 65%, which resulted in an enhancement in the overall performance by around 324.3%. In addition, Xu et al. [32]reported that the cooling fluid should not have volume flow rate of less than 100 L/h.m^2 in order to keep the thermal efficiency above 60%. Fig. 13 shows the daily energy output from the suggested PV/T system relative to the conventional stand-alone PV system.

Hou et al. [33] developed a technology that was based on micro heat pipe array (MHPA). A mathematical model was developed to analyze the new hybrid PV/T system with the new heat pipe array. Validating the numerical results with experimental data, the thermal efficiency of the new system, depending on the seasonal temperature

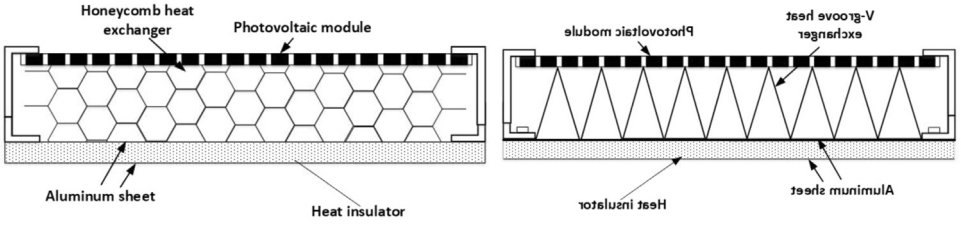


Fig. 7. PV/T collector with: (a) honeycomb heat exchanger, (b) V-groove heat exchanger (redrawn from [28]).

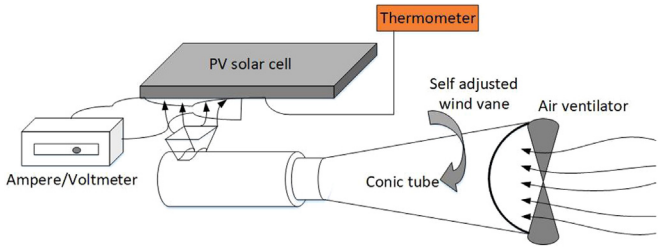


Fig. 8. A schematic diagram for the application of the wind tunnel cooling device with the PV system (redrawn from [29]).

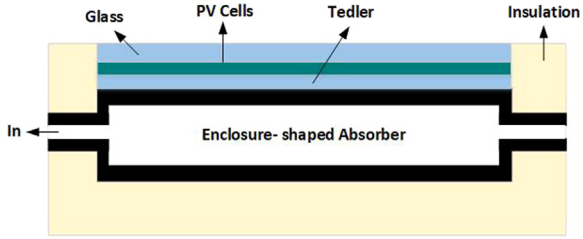


Fig. 9. Schematic diagram showing the enclosure-shaped absorber (redrawn from [30]).

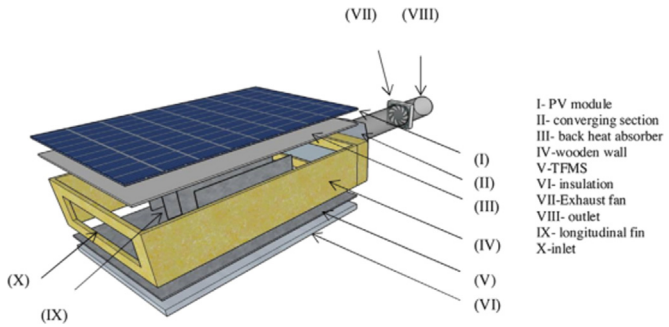


Fig. 10. Schematic diagram for the fins-attached air-cooled hybrid PV/T system (reprinted from [31], with permission from Elsevier).

reached around 40% in summer and went down to less than 20% in winter when the initial water temperature in both cases was at ambient temperature. The electrical efficiency was almost constant at around 13%. Jarimi et al. [34] carried out an experimental and developed a two dimensional numerical model to study the effect of using bi-fluid in cooling of hybrid PV/T system instead of single fluid on the performance of the system as shown in Fig. 14. They compared the performance under three different operating conditions of using only water, only air, or water and air together. The authors stated that the proposed design was very useful to be able to use different cooling modes by activating the required one. In addition, using bi-fluid has provided higher output power as well as higher energy saving relative to using air or water only that was 58.10% and 62.31% in respective manner,

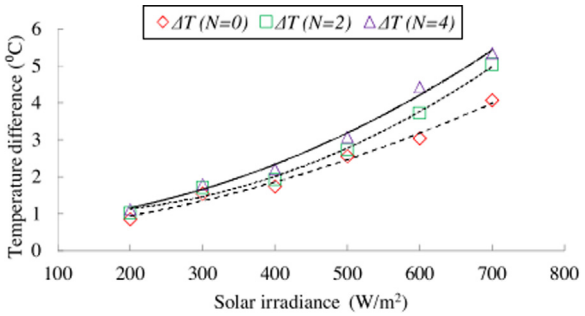


Fig. 11. Temperature difference accross the collector at different solar irradiance and number of fins (reprinted from [31], with permission from Elsevier).

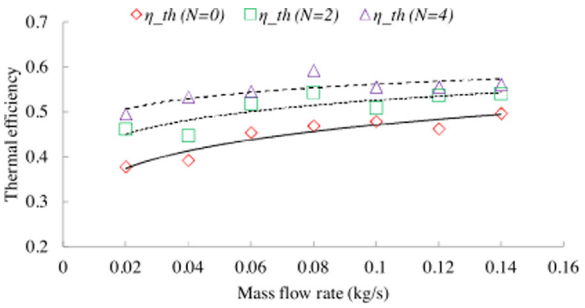


Fig. 12. Temperature difference accross the collector at different mass flow rates and number of fins (reprinted from [31], with permission from Elsevier).

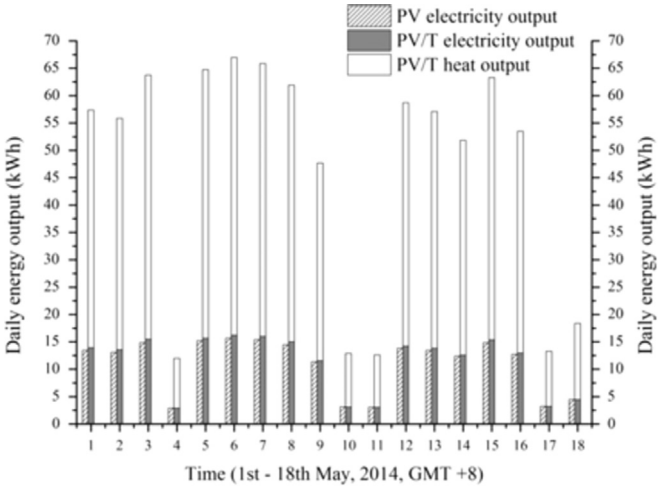


Fig. 13. Daily output energy from the hybrid PV/T system with super-thin absorber relative to stand-alone PV system (reprinted from [32], with permission from Elsevier).

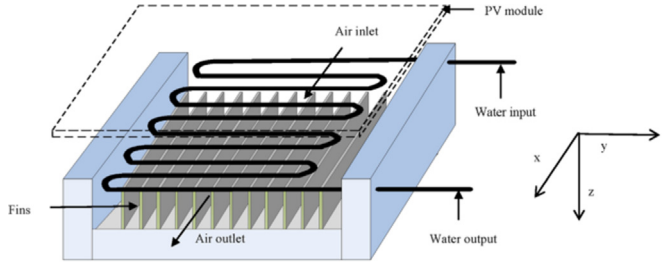


Fig. 14. Schematic diagram showing the bi-fluid cooling method of hybrid PV/T system (reprinted from [34], with permission from Elsevier).

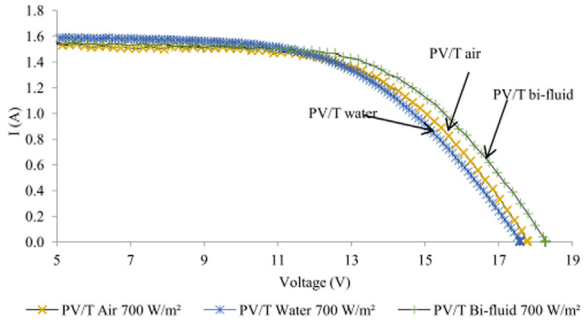


Fig. 15. Current-Voltage curve for the air, water, and bi-fluid cooled PV/T systems at constant irradiation and mass flow rates (reprinted from [34], with permission from Elsevier).

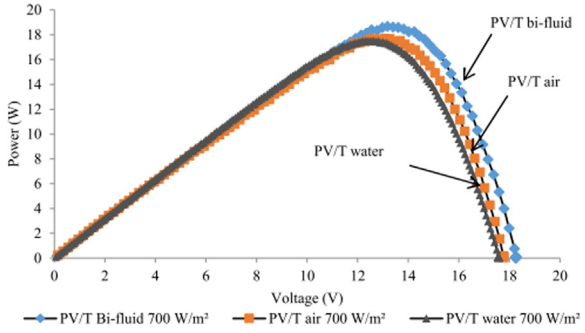


Fig. 16. Power-Voltage curve for the air, water, bi-fluid cooled PV/T systems at constant irradiation and mass flow rates (reprinted from [34], with permission from Elsevier).

Figs. 15 and 16. Table 2 summarizes the contributions and results of the previously mentioned studies.

Different studies had been conducted on the enhancement of the PV/T solar systems by using different designs of the system as well as the cooling system presented in the flow channels. Many factors had been studied: the channels arrangements, shape, thickness and cross section, integration of other components (e.g. turbine), use of fins and microchannel technology. Spiral flow channels and honeycomb cross-section had shown very good enhancement in the overall performance of PV/T system. Surrounding the cooling fluid by the absorber material and using fins were found as better than using only single bare layer. Thin channels had allowed for more heat transfer and better cooling for the PV cells, which in consequence resulted in high overall performance. Many studies still in a need to be conducted on how to integrate all the discussed techniques at the same time with optimization to reach the highest possible enhancement in hybrid PV/T performance.

6. Cooling fluids

PV panels usually benefits only from about 15–20% from the incident solar radiation and convert it to an electrical energy and the rest, after the radiation losses; is absorbed in the form of heat which ends up heating the PV cells and degrading their performance as shown in Fig. 17. Raising the temperature of the cells by about 1 °C above the normal operating temperature would lead to electrical efficiency degradation by about 0.45–0.65%. Chow [24] informed that based on the PV type, the degradation of the electrical efficiency as a result of temperature increase fall in the range of 0.25–0.5%/1 °C. Cooling of the PV became necessary in order to increase the lifetime of the cells [11] as well as to maintain the electrical efficiency of the PV at high level [35] which means reduction in the required PV modules' area. Many cooling technologies were studied in this field starting from the primitive methods like enhancing the heat losses from the surface by using materials that have high thermal conductivities up to using cooling fluids that should have the capability of decreasing temperature adequately to reach the optimum operating value. Cooling fluids used with the PV started by naturally moving air, going through forced convection air cooling, but air actually didn't present good heat transfer enhancement due to the low density, specific heat, and thermal conductivity as the maximum attained electrical and thermal efficiencies are 8% and 41% respectively [36,37].

Cooling is more effective by using water. The electrical and thermal efficiencies have increased to be 9.1% and 41.9% [39] respectively as a result of higher thermal properties as compared to air [37]. In addition, researchers have used refrigerants as cooling fluids for the PV panels. In this technology evaporation coils are installed beneath the PV cells to allow for low temperature refrigerant to pass through it which will be responsible in decreasing the PV cell temperature and as a result increasing the overall efficiency of the PV/T system [37]. There were some problems that resulted by the usage of such systems like the possibility for leakage of refrigerant. Using this method, electrical and thermal efficiencies reached 12% and 50% respectively [40]. The highly promising enhancement was achieved using nanofluids. Nanofluids are simply nanoparticles that are from metals or metal oxides in the micrometer or nanometer scale that are dispersed homogeneously in the base liquid. These nanoparticles have high thermal properties that allow it to be able to enhance the thermal properties and especially the thermal conductivity of the liquid once dispersed in the base fluid. Increasing thermal conductivity can result in reduction in operating cost and attaining higher thermal efficiency and better performance [41–43]. There was two ways of cooling either passive or active. For the passive cooling, there is no need for any pumping or compressing to circulate the cooling fluid and the fluid circulation occurs naturally due to gravity [11]. In contrast, pumping is needed in the active system. Hence, passive systems need less energy and low maintenance [9,17]. In addition, nanofluids are used as optical filters for the solar radiation. The objective of the optical filter is to absorb all the solar radiation that is not in the spectral range that corresponds to the working range of the PV cells. These filters have the advantage of absorbing over the unnecessary irradiation before reaching the cells, and hence reducing the temperature of the PV cells [14]. Examples of nanoparticles that can absorb at the visible range are: aluminum, copper, gold, lithium, palladium, platinum, and silver [14].

Arthur and Karim [43] proposed a comprehensive review on the thermo-physical and rheological properties of molten salt nanofluids. The experimental studies were conducted to investigate their effects on the solar thermal energy systems. They focused on nanofluids that can reach and withstand high temperatures. Base fluids that are capable of reaching high temperatures are synthetic oils and molten salts. Even at very small concentrations, dispersing nanoparticles in the base fluids will enhance the thermal conductivity and will result in an increase in fluid viscosity. They concluded that the best enhancement was obtained for MWCNT with 1 wt% loading. In addition, an enhancement of 10–30% in the specific heat was achieved without even taking the particle shape, type, or loading into consideration.

Nizetic et al. [10] provided an experimental as well as numerical

Table 2
Summary of the recent studies related to the design of PV/T system.

Year	Author	Contribution	Results
2014	Fudholi et al. [26]	Examined web flow absorber, direct flow absorber, and spiral flow absorber at different mass flow rates and solar irradiation	- As mass flow rate increase temperature decreases and electrical efficiency increases. - As irradiation increase, temperature and efficiency increase. - Higher thermal efficiency with direct web absorber. - Higher electrical efficiency with spiral flow absorber. - Highest overall performance with spiral flow absorber at 800 W/m^2.
2013	Hussain et al. [27]	Aluminum honeycomb heat exchanger below the PV panel.	- Thermal efficiency enhanced from 27% to 87%. - Negligible enhancement in electrical efficiency (Due to using air as cooling fluid).
2013	Yusof et al. [28]	Air cooled PV/T with either V-groove, honeycomb, or stainless steel wool,	- Honeycomb design has the best performance - Thermal and electrical efficiencies increases usually with mass flow rate. - Performance of stainless-steel wool is very close to honeycomb.
2014	Rahimi et al. [29]	Used a device based on a conic wind-collecting tunnel to cool the PV	- Electrical energy generated from the PV and the turbine. - The total output increased by about 36%
2011	Touafek et al. [30]	Used an enclosure-shaped absorber instead of absorber plate	- Overall efficiency was around 80% - Higher thermal performance if compared to the traditional single-layer absorber plate.
2016	Chandra et al. [31]	Attached thin rectangular fins in an air cooled PV/T system	- Overall performance has enhanced using fins. - Overall performance enhances as number of fins, air mass flow rate, and/or solar radiation intensity increases.
2015	Xu et al. [32]	Used super-thin absorber below the PV panels in a PV/T system	- Electrical efficiency has enhanced by 5% relative to stand-alone PV system. - Thermal efficiency of 65%, which can be maintained by volume flow rate more than 100 L/h.m^2.
2016	Hou et al. [33]	New PV/T based on micro heat pipe array (MHPA) technology	- Seasonal variable thermal efficiency (around 40% in summer and 20% in winter). - Almost constant electrical efficiency at about 13%.
2016	Jarimi et al. [34]	Used system of pipes and fins to apply bi-fluid cooling (water and air)	- Different cooling modes can be achieved. - Higher output power and energy savings relative to single-fluid cooling (air-cooling or water-cooling).

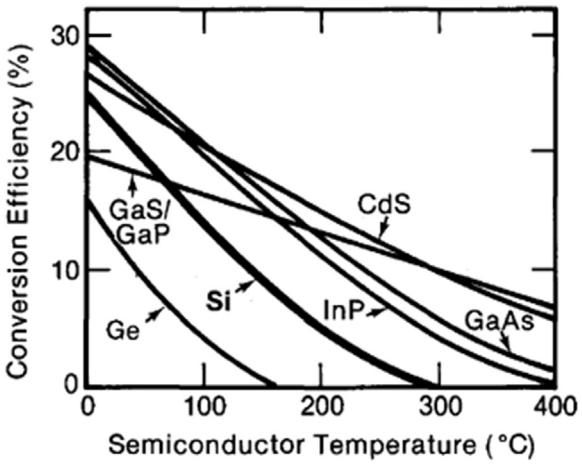


Fig. 17. Efficiency degradation with temperature for different PV cell materials [38].

investigation of the effect of backside air cooling in enhancing the performance of PV system. They reported that the backside suffers high temperatures in the range of 45–55 °C even with cooling. It was recorded that as irradiation increased and wind speed decreased, high temperatures were attained and vice versa. The big problem that was discovered was the flow separation again which occurred on the panel backside and resulted in thermal insulation and increase in the average PV cells' temperature and increased the degradation of electrical efficiency. They concluded that for a typical PV panels, the electrical efficiency degraded by about 2.5–4.5% due to flow separation leading to a result that the backsides should be redesigned to be more effective with cooling and therefore enhancing the electrical efficiency. Su et al. [44] proposed a study for the performance of PV/T with dual channels for different fluids. The performance analysis was done through comparison between four PV/T collectors with different fluids. The most efficient in both thermal and electrical efficiencies was the water-water cooled PV/T, while the outlet water temperature in the air-water cooled PV/T was the highest. They also found that for the water-water cooled design, increasing the mass flow rate of water as well as the height between the flow channels results in better overall efficiency. Hence, it can be said that using pure water or water-based nanofluids on the top of the PV cells can enhance the performance more than using air (Figs. 18-21).

Kourav et al. [35] proposed a review on the advancements and applications of nanotechnology in solar systems to attain better energy conversion efficiency. They noted that the existence of nanoparticles had an important effect in decreasing the PV cell average temperature if compared to that when using base fluid only. They concluded that the factors affect the cooling performance were mainly the channel geometry, nanofluid concentration and flowing regime (laminar or turbulent). Al-Waeli et al. [45] carried out an experimental study to show the effect of nanoparticles type and concentration on the performance of a nanofluid-based PV/T system. They aimed to get the highest possible thermal conductivity with the lowest possible increase in density and viscosity. They had studied different materials (Al_2O_3, CuO and SiC) at different volume concentrations (0.5%, 1%, 2%, 3% and 4%). Silicon carbide was found as the best from the stability and thermal conductivity points of view. They had found that higher concentrations led to higher output performance of the PV/T system, but lower fluid stability. In addition, lower PV temperature had been achieved using nanofluid as compared to the pure water. Al-Waeli et al. [46] experimentally evaluated the performance of SiC/water nanofluid-cooled PV/T system. Indoor and outdoor tests were done to accurately assess the performance. The density and viscosity of the nanofluid were increasing with nanoparticles concentration and decreasing with temperature. Thermal conductivity was increasing with both temperature and nanoparticles concentration. In addition, the stability of the nanofluid was degrading with time in the shape of reduction in thermal conductivity.

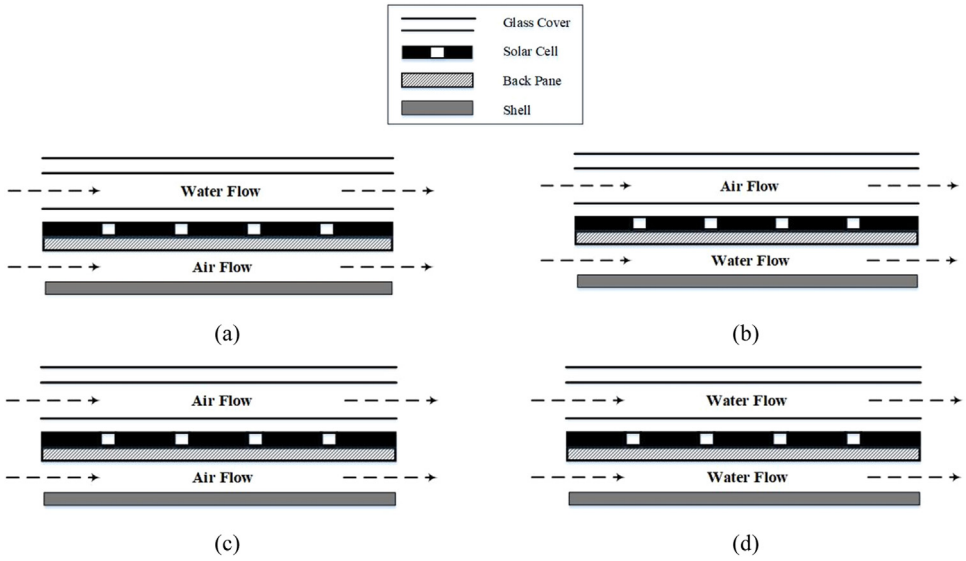


Fig. 18. The sectional view of the PV/T collector with dual channels for different fluids (reproduced from [44]).

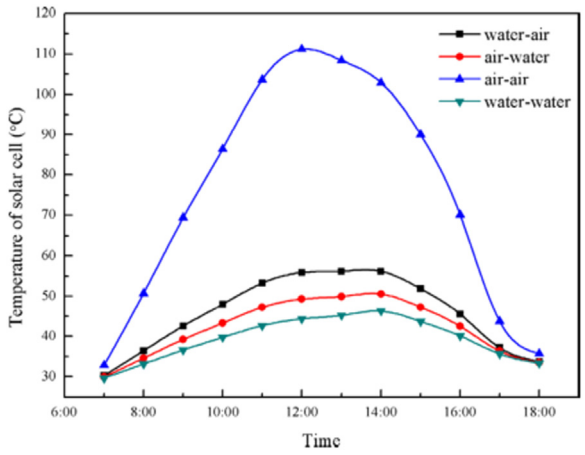


Fig. 19. The hourly variation of solar cell temperature (reprinted from [44], with permission from Elsevier).

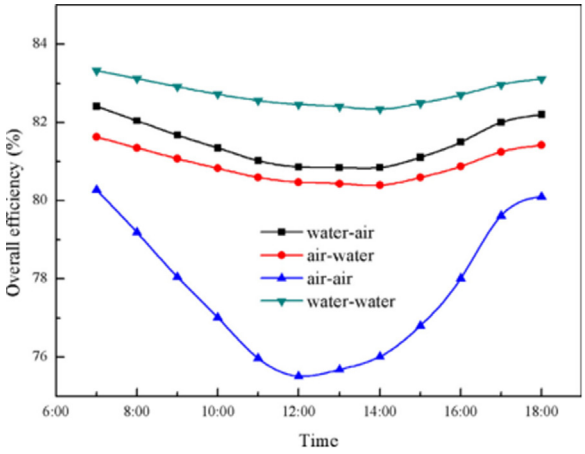


Fig. 20. The hourly variation of overall efficiency (reprinted from [44], with permission from Elsevier).

They recommended re-mixing the nanofluid each three weeks to ensure good stability and high thermal conductivity for the mixture. Hajjar et al. [47] proposed a study on the thermal conductivity enhancement

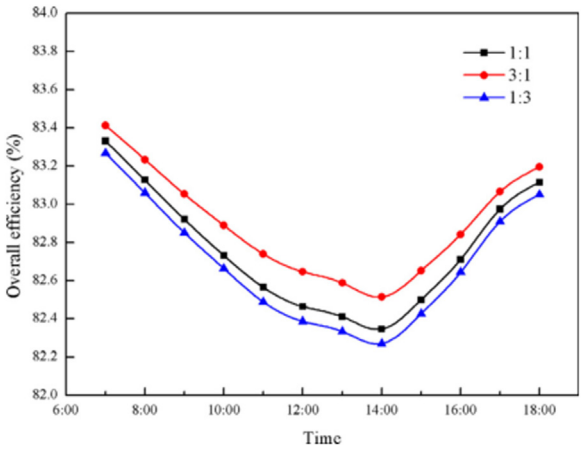


Fig. 21. The hourly variation of overall efficiency with different height of the pipes (reprinted from [44], with permission from Elsevier).

as a result of using graphene oxide nanofluids. Graphene oxides sheets showed good compatibility with water that they are hydrophilic surfaces, so dispersion in water do not need any surfactants. Ultra-sonication was enough. This compatibility helped in enhancing the thermal properties of the fluid. Through all types of GO/water nanofluids, GO-5 had the best enhancement in thermal conductivity relative to that of water (about 47.54% more than water at 40 °C). In addition, in regards to the optical properties, they examined high absorptance for the graphene oxide nanofluid in the spectral range of solar radiation. Vakili et al. [48] and his group examined the effect of using graphene nanoplatelets in enhancing the heat transfer in solar collectors, when dispersed in a deionized water as a base fluid. The authors tried different concentration in weight fractions of 0.0005, 0.001, and 0.005, and different mass flow rates of 0.0075, 0.015, and 0.225 kg/s. The efficiency was increased with direct proportion to the nanoparticles concentration and the maximum value was at a mass flow rate of 0.015 kg/s. Finally, the efficiency was enhanced at concentrations of 0.0005, 0.001, and 0.005 to be 83.5%, 89.7% and 93.2% respectively, whereas that for base fluid was 70%. Vincely and Natarajan [49] studied the usage of graphene oxide nanofluid as working fluid with the flat plate collector. The fluid was examined under different nanoparticles concentrations and nanofluid flow rates. It can be said that as the volume concentration increased, the heat transfer coefficient increased. They

Table 3
Summary of the recent studies related to the cooling fluids.

Year	Author	Fluid	Results
2016	Nizetic et al. [10]	Air from the backside of PV	- High temperature of backside even with cooling (45–55 °C). - Flow separation on the backside results in thermal insulation and increase in temperature. - Electrical efficiency will be degraded by 2.5–4.5%
2003	Zondag et al. [23]	Water on top of PV	- Thermal efficiency enhanced and amount of heat absorbed by the water has increased as compared to bottom cooling.
2017	Odeh and Behnia [11]	Spill water from front side of PV	- Electric power has enhanced by 10.3%. - Cell temperature has decreased and incident radiation has increased due to refractions in water.
2016	Su et al. [44]	Different fluid combinations from top and bottom sides of PV panel	- Highest electrical and thermal efficiencies using water-water cooling from the top-bottom sides of the PV panel. - Highest outlet water with air-water cooled system. - Increase of mass flow rate enhances the overall performance.
2017	Al-Waeli et al. [45]	Al_2O_3/water, CuO/water and SiC/water nanofluids	- Silicon carbide showed the best stability and highest thermal conductivity. - Higher nanoparticles concentrations resulted in higher performance of the PV/T, but lower fluid stability. - Lower PV temperature had been achieved using nanofluid as compared to pure water.
2014	Hajjar et al. [47]	GO/water nanofluid using GO sheets	- Good compatibility between GO sheets and water so good dispersion in water. - Thermal conductivity has enhanced by 47.54% relative to water.
2016	Vakili et al. [48]	Graphene nanoplatelets/deionized water nanofluid	- Thermal efficiency increases with nanoparticles concentration. - Thermal efficiency increases with mass flow rate until certain limit and start to decrease.
2016	Vincely and Natarajan [49]	GO/water nanofluid	- Heat transfer coefficient increases with nanoparticles concentration. - Viscosity slightly increases with nanoparticles concentration. - Thermal efficiency decreases as inlet fluid temperature increases. - Increase of concentration may lead to decrease in thermal efficiency if heat losses increased.
2017	Yazdanifard et al. [50]	Al_2O_3 and TiO_2 dispersed in either water or EG-water mixture	- Al_2O_3 showed better performance than TiO_2. - Insignificant effect of the nanoparticles size. - Water showed better performance on the energy and exergy levels than EG-water mixture base fluid. - Addition of nanoparticles were more effective in laminar flow rather than turbulent flow. - Increasing of concentration at constant mass flow rate showed enhancement in the outlet fluid temperature. - Reduction in PV temperature in laminar regime
2015	Qiu et al. [37]	Micro-encapsulated PCM slurry in water or EG	- High heat capacity and thermal conductivity. - No effect on temperature or efficiencies with concentration less than about 10 wt%. - PV temperature decrease at higher concentrations and Reynolds number.

found that viscosity increased slightly with increasing the volume concentration, and as temperature increased, viscosity decreased and the pumping power decreased as well. The group noticed that increasing the mass flow rate increased the thermal efficiency. In contrast, increasing the inlet water temperature decreased the thermal efficiency. In addition, they found that for the same flow rate, using nanofluid showed better performance at low inlet temperatures as compared to the base fluid, while the base fluid was better at high temperatures. They found also that at constant mass flow rate, increasing the mass concentration would not always lead to an increase in the efficiency; the reverse may occur, because of high heat losses. Furthermore, they found that the maximum heat gain was attained at mass concentration of 0.01 and mass flow rate of 0.0167 kg/s. They concluded that the heat transfer coefficient increased as Reynolds number and the mass concentration increased. Yazdanifard et al. [50] demonstrated a review for the effect of nanofluids on the performance of PV/T systems. They had studied the effect of different nanofluid parameters such as type of base fluid and nanoparticles, volume fraction and nanoparticles size. For the nanoparticles type, alumina was found as showing better performance than Titanium Oxide. The dependence of the results on the nanoparticles size was insignificant. For the base fluid type, water showed better performance on the energy and exergy levels than ethylene glycol-water mixture base fluid. They had found that addition of nanoparticles to base fluid was more effective in laminar flow rather than turbulent flow. Increasing of nanoparticles concentration at constant mass flow rate showed enhancement in the outlet fluid temperature. In addition, reduction in PV temperature in laminar regime had been achieved, while it increased slightly in turbulent one.

Qiu et al. [37] presented a model for using the microencapsulated PCM slurry in enhancing the performance of PV/T solar system. A validated mathematical model and computer program were proposed. Effects of the serpentine size, concentration ratio and Reynolds number were discussed. The authors used amorphous silicon PV cell material as it offered lower electrical efficiency degradation with temperature increase than the crystalline silicon cells (0.25%/1 °C and 0.5%/1 °C respectively [51,52]). As PCM materials suffer mainly for very low thermal conductivity and hence low heat transfer from the PV. The group used encapsulated PCM particles in very small sizes (1–100 μm) with polymer shells and ended up with what is called MPCM. Dispersing these particles in heat carrier fluid like water or ethylene glycol forming a slurry with higher thermal conductivity and heat capacity. They reported that up to about 10 wt% mass concentration, the temperature of the PV cells, electrical, and thermal efficiency remain almost not changed and as Reynolds number and MPCM mass fraction increased the PV cells' temperature decreased. In contrast, as Reynolds number increased, and MPCM mass fraction decreased, pressure drop decreased. In their paper, they defined two efficiencies describing the total performance; the overall efficiency, which is the addition of thermal and electrical ones, and the net efficiency, which took into consideration the power losses due to pumping. Increase of nanoparticles concentration led to an enhancement in the heat transfer but at the same time, the fluid viscosity increased, resulting in decreasing Reynolds number, increasing pressure drop and pumping power and as

a result decreased the net efficiency, so an optimization became necessary.

Location of the working fluid in the PV/T is very important; it can be placed on the front side of the PV panel and/or its backside. Each location has its own advantages and drawbacks. The back surface of a PV panel has the highest temperature due to the high thermal conductivity of the silicon cell material [53], hence the main cooling should be from the backside. Zondag et al. [23] found that using the working fluid on the top of the PV enhanced the thermal efficiency and increased the amount of heat absorbed by the fluid, but on the other hand it decreased the transmissivity to the solar incident radiation and as a result decreased the electrical efficiency, and vice versa. In order not to prevent the incident solar radiation from reaching the PV and at the same time keep the thermal efficiency at high relative values, they found that the most suitable PV/T design was achieved when the channels were located beneath transparent-PV panels. Actually, they solved the problem partially from the side of thermal collected energy. This problem is almost solved when the idea of spectrally-selective-nanofluid-optical filters arise. In optical filters, the nanofluid presents in between the solar incident radiation and the PV panels. Odeh and Behnia [11] studied the cooling of PV solar collector for irrigation purposes using water. By spilling water by gravity on the front surface of the PV panels, they found that cooling from the front side enhanced the electric power by around 10.3%. Actually, they found there is more than one advantage from using water on the front surface of the PV; decreasing the cells temperature and increasing in incident radiation due to refractions by water. They realized an enhancement in the system output. The cooling efficiency increased because of direct contact between water and the PV panel and the incident radiation on the PV had increased due to refraction of the solar beam inside the water layer. Table 3 summarizes the contributions and results of the previously mentioned studies.

There is no doubt that enhancement of the cooling and heat transfer in PV/T systems became a necessity. Nanofluid had proved itself as a step forward in the cooling effectiveness. Carbon-based nanofluids had shown the best enhancement in thermal properties among all the other nanofluids. Extra work still needed to come up with the best choice of carbon-based nanofluids that will achieve stability, economic visibility and highest enhancement in the PV/T systems output.

7. Optical filtration

Hjerrild et al. [14] discussed another approach in using of nanofluids in cooling of PV/T. They studied the concept of optical filters using nanofluid composed of mixture of particles of Ag-SiO_2 and CNT. Ag-SiO_2 was responsible for absorbing the radiation corresponding to the visible light, while CNT was used to enhance the absorption. They found that the nanoparticles need to be coated sometimes in order to withstand higher temperatures. Silver nanoparticles can be said to be the strongest nanoparticles that when suspended in a base fluid will have the capability of absorbing radiation in the range of 300–1200 nm [54]. They proposed that in order to decrease the scattering by the nanoparticles, and as a result decrease energy lost; the volume concentration of the nanoparticles should be reduced ($< 0.6\%$) and the particles diameter should be less than 50 nm. Due to that the shape and size of the nanoparticles are important and because of the high surface area to volume ratio, their melting temperatures become lower than that for bulk material and at the UV range with high temperature. This may cause the Ag nanoparticles to change its shape and therefore the absorption peak. Further, they coated the particles by a silica shell in order to maintain the shape suitable for absorption. They concluded that silver particles and presence of CNT in the suspension highly enhanced the absorption of solar radiation and as a result increased the thermal efficiency of the PV/T.

Hassani et al. [55,56] used spectrally-selective nanofluids to have the best transmitting properties in the useful spectral range for PV electrical conversion, and the best absorption properties in the rest of spectrum. In these studies, the nanofluid-optical filter did not prevent solar radiation from reaching the PV panels, and at the same time, it gets the most benefit from the solar thermal energy. Their nanofilter was able to absorb practically all the UV and the IR spectra, which are not suitable for PV electrical conversion. Hassani et al. [56] found that using silver/water nanofluid is better than only water, as water alone is not capable of absorbing except the IR spectra. Crisostomo et al. [57] experimentally studied the effect of different types of nanofluids to work as optical filter with a concentrated type hybrid PV/T collector. They have mentioned that these nanofluids could be used efficiently as a bandpass optical filters based on their capability of beam splitting. Comparing the output of the hybrid system to the electrical output from a conventional stand-alone PV system, using only water as heat transfer fluid (HTF) would be able to deliver extra heat as compared to standalone system as it absorbed radiation at long wavelengths. Moreover, using nanofluids allowed the absorption of radiation from shorter wavelengths and to reach higher stagnation temperatures. By dispersing silver-silica core-shell nanodiscs in water, the overall output had been enhanced by around 11% as compared to water as a HTF.

Jing and Song [58] investigated a study on the optical properties of the nanofluids and the effects of the size and distribution of the nanoparticles in enhancing these optical specs. Jing and Song [58] mentioned that due to the high surface energy of the nanoparticles even with small diameters, they tend to agglomerate. This agglomeration affected the outcome optical properties. To date still not enough studies have covered these issues. Jing and Song [58] have gone through different models simulating the motion of the nanoparticles and the effect of agglomeration on the optical properties. Meanwhile, agglomeration is a complicated process that needs high-powered hardware to simulate a large system like nanofluid system. In addition, Jing and Song [58] mentioned that the models describing the agglomeration of nanoparticles could be used after defining the agglomerated detailed structure, which is not an easy matter. Lastly, they had found that absorption could be increased by increasing the volume fraction, which on the other side may negatively affect the stability of the nanofluid. Gorji et al. [59] studied the optical properties of a functionalized carbon nanotube dispersed in deionized water. The absorption coefficient of the prepared nanofluid was quantitatively analyzed and compared to pure water. The analysis was done at room temperature and different higher temperatures. Gorji et al. [59]found that dispersing CNT in the water increased the absorption coefficient and it decreased as temperature increase as shown in Fig. 22. In addition, they had found that functionalized CNT was more stable than pristine CNT when dispersed in deionized water. Functionalizing of the CNT was required to solve the problem of hydrophobicity of the pristine CNT. It was concluded

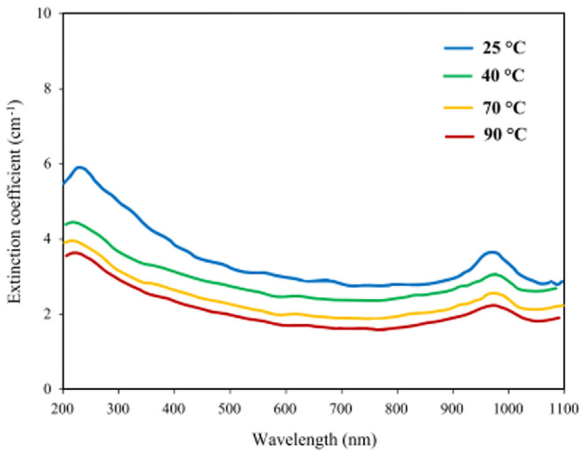
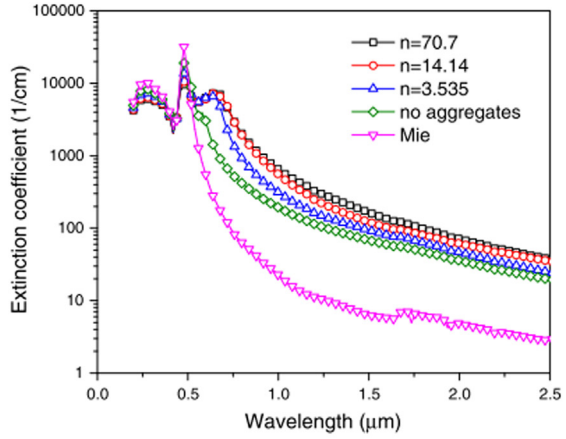
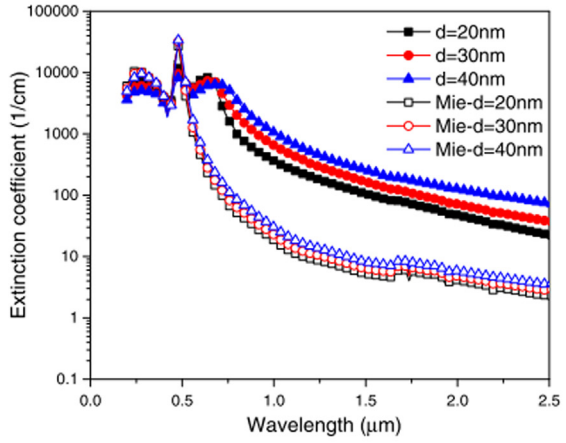


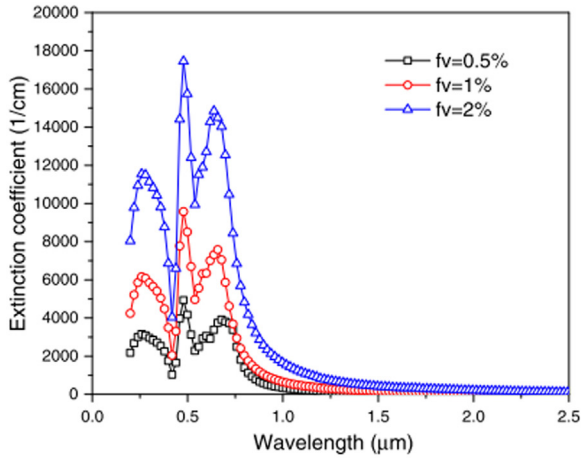
Fig. 22. Decrease of extinction coefficient with increasing temperature of nanofluid (reprinted from [59], with permission from Elsevier).



(a)



(b)



(c)

Fig. 23. Extinction coefficient dependency on: (a) agglomeration, (b) particle size, and (c) volume fraction (reprinted from [60], with permission from Elsevier).

that functionalized CNT could enhance the absorption coefficient while have a stable nanofluid.

Du and Tang [60] studied the optical properties of Ag/water nanofluid under the effect of agglomeration using theoretical approach. Their study involved the use of Diffusion Limited Cluster Aggregation (DLCA) simulation and Generalized Multi-particle Mie solution (GMM) method. They had found that agglomeration could increase the extinction coefficient at high wavelengths while it decreased the absorption at short wavelengths, Fig. 23a. In addition and as a result, extinction coefficient was enhanced for high particle sizes at high wavelengths and was decreased at short wavelengths, Fig. 23-b. Furthermore, Du and Tang [60] have found that as the volume fraction increased the agglomeration would have better effect on increasing the absorption coefficient of the nanofluid, Fig. 23-c.

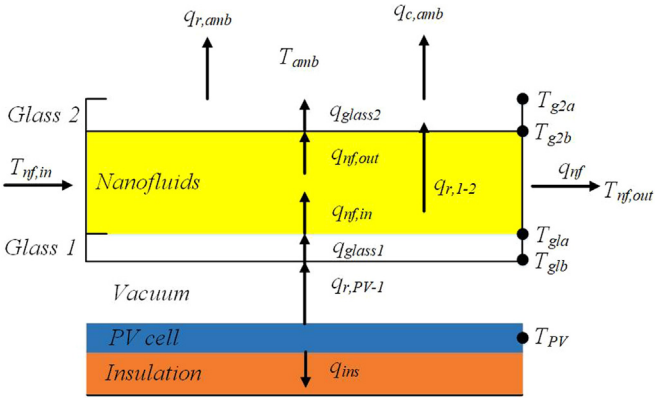


Fig. 24. Schematic diagram for hybrid PV/T using optical filtration nanofluid (redrawn from [61]).

Jin and Jing [61] proposed a novel magnetic electrolyte nanofluid (ENF) to be used as optical filter with hybrid PV/T collectors as depicted in Fig. 24. The nanofluid was composed of magnetic Fe3O4 nanoparticles dispersed in a 50% water/50% EG solutions with the addition of methylene blue (MB) or copper sulfate (CS). The optical filter was tested with two different PV cells: Si and lnGaP. Testing the proposed magnetic optical filter against traditional optical filters with the two PV/T system, they had found that it had good stability. Higher optical efficiency was presented for ENF as compared to core/shell nanoparticle traditional filters with smaller thickness used. In addition, ENF showed higher thermal conductivity as compared to base fluid in small temperature ranges. They had reported that ENF could be considered as an efficient and economical liquid optical filter.

Crisostomo et al. [62] proposed a numerical study discussing the effect of nanoparticles material, size, and shape on the optical properties of the nanofluid suspension. They had suspended a core-shell Ag-SiO2 in water to filtrate the light spectrum fall into a PV panel and had ensured that the optical transmission of the suspension lied in the ideal transmission range for the PV cells. The results from the model were validated with an experimental system and it was found that the weighted average energy output from the PV system could be increased by around 12% if compared to a conventional PV system.

Hjerrild et al. [14] investigated experimentally and numerically the usage of optical filters in between a concentrated light source and PV cells as shown in Fig. 25. The optical filter nanofluid composed of core-shell Ag-SiO2 nanodiscs and CNT dispersed in water. As previously mentioned, the suspension was synthesized to absorb the majority of the incident visible light while just transmitting the spectral range that is corresponding to the electrical conversion of the cells. They had found that the silver nanoparticles provided high absorption. The usage of silica shell prevented any thermal or UV degradation. Alternatively, addition of CNT helped in enhancing the absorption in the UV range in addition to enhancing the heating rate of the nanofluid, (see Fig. 26). Economically, they had proved that adding extra cost of less than 1$/L for nanoparticles, a promising enhancement could be achieved in both the electrical and thermal outputs relative to optical filtration using traditional base fluid.

Table 4 summarizes the previous studies discussed the optical filtration topic.

Usage of the nanofluid as an optical filter with absorption systems

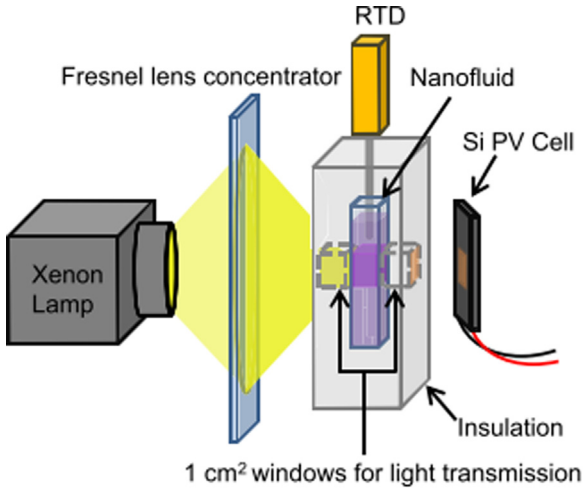


Fig. 25. Schematic diagram of the experimental setup (reprinted from [14], with permission from Elsevier).

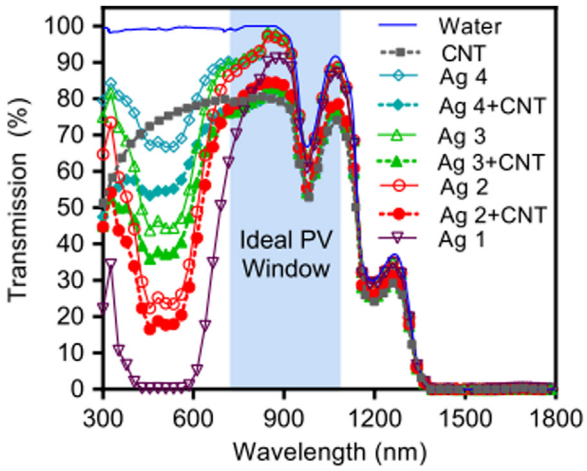


Fig. 26. Transmission spectrum of different tested optical filtration nanofluids relative to water (reprinted from [14], with permission from Elsevier).

had been studied many times in the literature. On the contrary, integration of nanofluid as an optical filter with the hybrid PV/T systems is considered as a new hot area for research that still in a need for extra work to study all the parameters controlling the behavior of the optical fluid.

8. Storage

As a heat removal medium, both water and air have been utilized in different applications. A big drawback of the solar energy its intermittency and as a result high mis-match between the supply and demand. In the night time, the heat demand is maximum, while the solar radiation intensity is minimal or even zero [63], so thermal energy storage become a need. Some requirements need to be met to obtain an efficient thermal energy storage system. It should have sufficient storage volume, high efficiency, quick charging/discharging capability, and to be economically effective and free of hazardous materials [64]. Regarding the storage medium, water and air are not suitable to store energy in the periods of high insolation to the periods of zero insolation. Phase change materials were found to be the best for this purpose.

Phase Change Materials (PCMs) commonly refers to those materials, which use the solid-liquid phase change process to absorb and then release heat loads at constant temperature [37,65]. PCMs were preferred to work in the solid-liquid phase instead of liquid-vapor phase due to the large density changes than occurs in the later one [66]. Consequently, with solar radiation, PCM is heated sensibly until the melting/solidification temperature is reached, and with more heating the PCM progressively melts as a result of absorbing latent heat, and vice versa for the heat release time. Cabeza et al. [67] classified the PCMs based on their melting temperatures and enthalpies as shown in Fig. 27.

Availability of PCM should be in a system that has high temperatures at the daytime needed for melting of PCM and resulting in high heat extraction rate, and high temperature drop. Moreover, this system should have low temperatures at the night time, so higher PCM solidification rate is attainable for regeneration of PCM fully for the next day absorption [3].

Normally, the phase change occurs at a temperature range and duration that are mainly dependent on the mass, thermal conductivity, and heat capacity of the material [2]. Due to the capability of storing and releasing high amounts of thermal energy with small corresponding volume with nonsignificant variation in temperature, PCMs have been widely used in thermal storage [65,68–70]. Usage of PCM helped in increasing the time required for the PV surface to reach a certain temperature so keep it away from the problems results from high temperatures [71]. Ideal PCM should have certain properties like high thermal conductivity, high latent heat of fusion, suitable practical phase-transition temperature range [72], and high density with low volume change during change of phase. Also, it should be chemically stable along period of time, non-toxic, and non-hazardous [73]. PCMs are mainly classified as organic materials (e.g., paraffin and non-paraffin) and inorganic materials (e.g., salt hydrates, eutectics pentaerythiol and pentaglycerine) with both have melting temperatures in the range from 15° to 30°C [37]. Paraffin waxes which are considered as the commonly used PCMs for technical applications have found to match the majority of the required properties, but they have problems with low thermal conductivity [72] and high volume change during melting [73]. Thermal stability is very important as well for these systems that experiences continuous heating, which is highly satisfied using organic PCMs [9,74]. Indeed, organic PCMs were found to behave better than the inorganic types but with the disadvantages of low thermal conductivity and lower phase change enthalpy [75]. Zalba et al. [75] summarized the pros and cons of organic and inorganic PCM as shown in Table 5. Many researches have been proposed with different enhancement techniques for the improvement of the thermal conductivity of the PCMs.

Browne et al. [9] discussed the use of phase change materials in the thermal management of PV, CPV and BIPV systems to regulate the temperature of the cells especially that made from crystalline silicon. They notified that as solar irradiation increases, the need for a PCM material that works at higher temperatures increases, Fig. 28. Reviewing different literatures, they found that the PCM melting temperature for thermal management of PV ranges between 18.7 up to 56 °C. They concluded that efficient re-solidification at nighttime was a very important parameter in enhancing PV performance when integrated with PCM. In addition, after reviewing many systems, they concluded that using PCM with PV was capable of delaying the temperature increase of PV, so enhancing the overall performance.

Su et al. [76] investigated a comparative study on the benefits behind using PCM in cooling the hybrid PV/T solar collectors. The effect of PCM position on the surface temperature, outlet temperature and the efficiencies were studied as well as the effect of the thickness of the PCM layer. Locating the PCM layer on the upper side of the duct gave higher overall efficiency because of improving both the electrical and thermal performances as depicted in Fig. 29 and 30. They found also, that for a PV/T with PCM layer on the top surface, a 3 cm thickness layer was excellent in both thermal and electrical efficiencies.

Malvi et al. [77] proposed an energy balance and simulation analysis for the combined PV/T that used PCM as storage material. They had studied the effect of PCM thermos-physical properties and the

Table 4
Summary of the recent studies related to Optical filtration nanofluid.

Year	Author	Optical nanofluid	Results
2016	Hassani et al. [55,56]	Ag/water nanofluid	- -Nanofluid did not prevent solar radiation from reaching PV. - Nanofluid absorbed the IR and UV radiations. - Using water alone would absorb only in IR spectra.
2015	Crisostomo et al. [57]	Water alone and Ag-SiO_2 core-shell nanodiscs/water nanofluid	- Nanofluids could be used as bandpass optical filters based on beam splitting capability. - Water alone could absorb radiation at long wavelengths only. - Using Ag-Si/water could absorb radiation at long and short wavelengths and enhanced the output relative to water by about 11%
2017	Jing and Song [58]	–	- Agglomeration of nanoparticles affected its optical properties. - Absorption has increased with volume fraction but it affected the stability and transmissivity of nanofluid.
2015	Gorji et al. [59]	CNT/deionized water	- Dispersion of CNT in water increases the absorption coefficient. - Absorption decreases with high temperature. - Functionalized CNT was more stable than pristine CNT when dispersed in water.
2015	Du and Tang [60]	Ag/water	- Agglomeration of nanoparticles increased and decreased the extinction coefficient at long and short wavelengths respectively. - Increase of particles size do the same effect as agglomeration.
2017	Jin and Jing [61]	Fe_3O_4/50%water + 50%EG with addition of methylene blue (MB) or copper sulfate (CS)	- Good stability of the nanofluid. - High optical efficiency as compared to core-shell nanoparticle traditional filter. - Efficient and economical optical filter.
2017	Crisostomo et al. [62]	Core-shell Ag-SiO2/ water nanofluid	- Enhancement by about 12% in the weighted average energy output from the PV if compared to conventional PV system.
2016	Hjerrild et al. [14]	Core-shell Ag-SiO_2 nanodiscs + CNT/ water	- Ag nanoparticles had high absorption in UV region and addition of CNT enhances the absorption. - CNT enhanced the heat rate of nanofluid. - Extra cost of less than 1$/litre can enhance the overall performance as compared to optical filtration based on only base fluid.

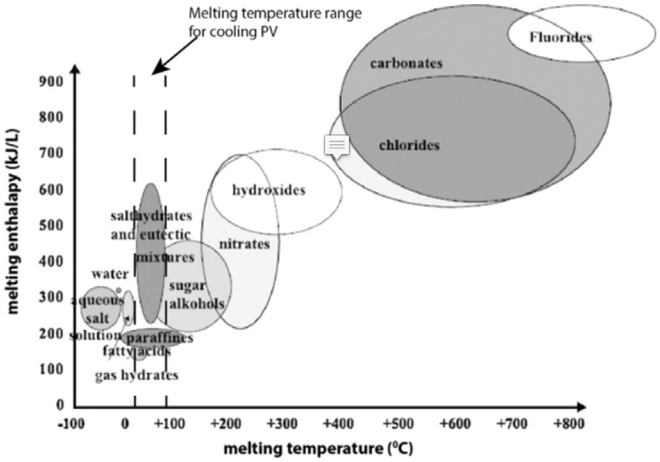


Fig. 27. PCM classification based on melting temperatures and melting enthalpy (reprinted from [67], with permission from Elsevier).

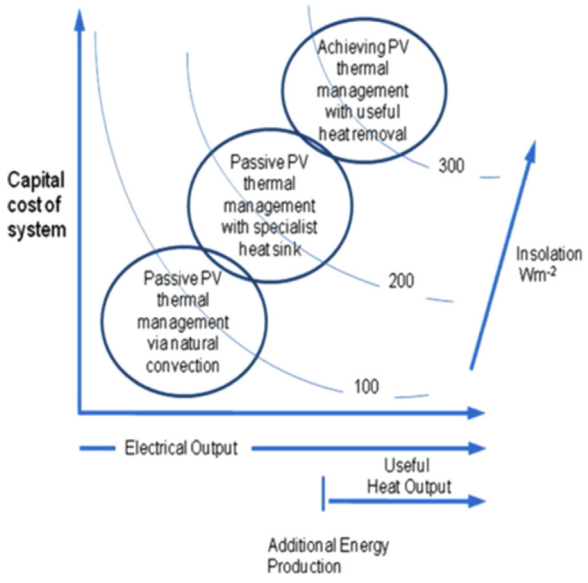


Fig. 28. Illustrative ranges for thermal management and heat removal in a PV system (reprinted from [9], with permission from Elsevier).

Table 5
Advantages and disadvantages of organic and inorganic PCM for thermal storage [75].

	Organic	Inorganic
Advantages	- No corrosives. - Low or none undercooling. - Chemical and thermal stability.	- Greater phase change enthalpy.
Disadvantages	- Lower phase change enthalpy - Low thermal conductivity - Inflammability	- Undercooling - Corrosion - Phase separation - Phase segregation, lack of thermal stability

cooling water behavior. They found that the performance of the PV could be enhanced by about 9% for an average temperature rise of 20, when the suitable PCM was used. They found that the increase in electrical performance would be accompanied with reduction in thermal behavior except if a certain optimization was achieved. They suggested that the performance of the PV/T system would be improved if the PCM material was developed to attain tunable melting points and high thermal conductivity.

Colla et al. [73] discussed the feasibility of using alumina and carbon black nanoparticles to enhance the thermal properties of pure paraffin waxes to obtain a new class of PCMs, called Nano-PCMs in order to enhance the heat storage capabilities. Two PCM materials were examined (RT20 and RT25) with the addition of nanoparticles (Carbon black and Aluminum oxide). They had noticed significant enhancement of the thermal conductivity after the addition of CB nanoparticles to RT20 and RT25 PCMs. The highest latent heat was found for RT25/

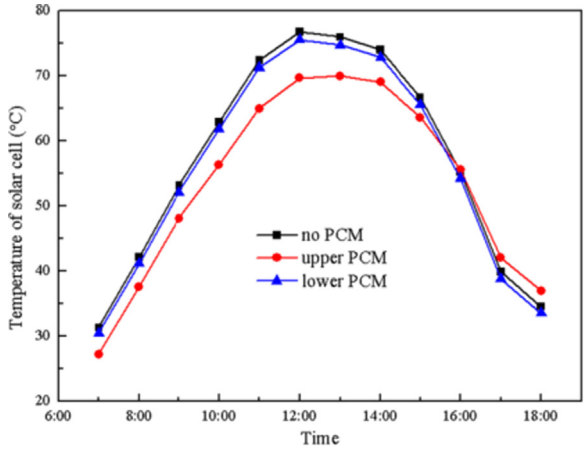


Fig. 29. The hourly variation of solar cell temperature (reprinted from [76], with permission from Elsevier).

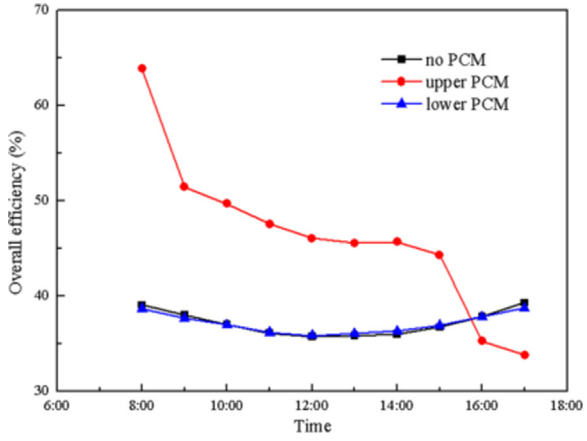


Fig. 30. The hourly variation of overall efficiency (reprinted from [76], with permission from Elsevier).

Al2O3 and the lowest for RT25/CB. Addition of Al_2O_3 had enhanced the latent heat, while addition of CB led to a degradation in the latent heat of the mixture as compared to the pure PCM materials. In addition, increasing the melting temperature of RE25 contributed in the latent heat enhancement. On the other side, Carbon black nanoparticles' addition enhanced the thermal conductivity while alumina did not have the same effect. Colla et al. [73] concluded that the addition of alumina could make an enhancement in the latent heat while CB would be able to make an enhancement in the thermal conductivity. Browne et al. [2] proposed a study on the utility of PCM for heat storage and water preheating in PV/T systems. The study was on four configurations; PV/T with PCM surrounding pipe network, PV/T, PV with container for storage, and PV only. The measurements were made for more than two days to ensure accuracy of results. As compared to the system without PCM, they had found that using PCM, the heat extraction process from the PV was improved by about seven times. Huang et al. [78] carried out an experimental study on a finned PV/PCM system to evaluate the effects of convection and crystalline segregation in a PCM on the thermal efficiency. Utilization of fins in the container of the PCM enhanced the temperature regulation because of increasing natural convection, which would affect the melting and solidification of PCM.

Hasan et al. [79] proposed a study on 4 different-cell-size PV-PCM systems using 5 different PCM types with melting temperatures in the range of 21–29 °C and heat of fusion between 140 and 213 kJ/kg to find the optimum PCM material and optimum cell size in order to attain the highest possible PV cooling for BIPV. The thermal performance presented high dependency on the thermal mass of the PCM and the

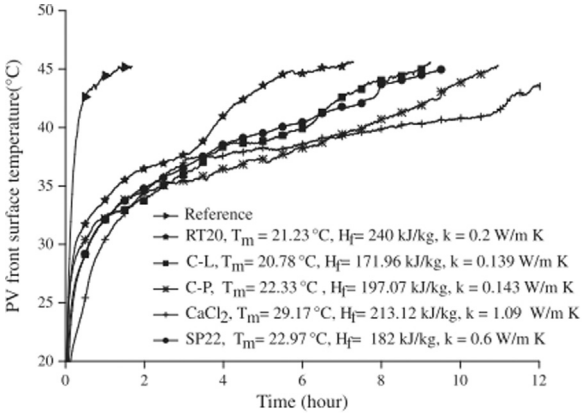


Fig. 31. Temperature evolution on PV front surface in system A at an insolation of 500 W/m2 and ambient temperature of 20 ± 1 °C (reprinted from [79], with permission from Elsevier).

thermal conductivity of the PV and PV-PCM designs. For specific solar radiation, changing the PCM material had allowed for more gradual or slow build-up for the temperature with time, Fig. 31. Regarding PCM types, the salt hydrate $CaCl_2$ presented the highest reduction of PV temperature for relatively long times at different irradiance levels. In addition, system A was the best design in terms of the highest temperature reduction. Hasan et al. [3] studied the effect of using different PCMs in regulating the temperature increase of the PV cells. In addition, they had studied the effect of PCM location on the possible attainable temperature drop. The study had been made in two different-climate locations. The properties of the PCMs selected were higher heat capacity and thermal expansion for Eutectic of capric–palmitic acid (PCM1) and higher thermal conductivity and density for Calcium chloride hexahydrate $CaCl_2$.6H2O (PCM2). $CaCl_2$.6H2O had achieved more temperature drop by about 3–4 °C and higher power saving by about 3% than capric–palmitic acid in the hot-climate locations.

They concluded that PV-PCM2 has the lower and more stable temperature fluctuation as well as higher open circuit voltage and that the maximum power and voltage improvements as a result of using PCM with PV was achieved at places encounter hot weather. Hasan et al. [80] examined five different types of PCMs at different thicknesses and different solar radiation intensities. They had reported that to maintain 10 °C of temperature reduction for the longest period, salt hydrate PCM with 50 mm thickness would be the best choice. On the other side, they reported that usage of salt hydrate might not be practical due to problems with thermal stability over long periods of operation. Hasan et al. [79] and Huang et al. [78] reported that the climatic weather affected the performance of the PCM attached with the PV. For hot weathers, ambient temperature was high and high heat was gained by the PCM at the daytime. During nighttime, heat extraction by PCM was not easily possible due to the high temperatures at the night, so solidification rate had decreased and resulted in incomplete PCM regeneration for the next day heat absorption. For cold weathers, lower ambient temperatures during daytime resulted in lower temperatures on the PV panels and low heat loads. In nighttime and using the lower temperatures, higher PCM solidification rate was attainable for regeneration of PCM fully for the next day. Atkin and Farid [17] proposed a study on the effect of using PCM infused graphite and aluminum fins on the performance of PV cells. They had studied four different models; PV only, PV with PCM infused graphite, PV with aluminum fins, and PV with PCM infused graphite and aluminum fins. The last design was the best and the overall efficiency increased by about 13%. In addition, the thermal conductivity had increased from 0.25 to 16.6 W/m.K by infusion of PCM into graphite.

As PCM originally has high latent heat storage advantage, but low thermal property, incorporation of nanomaterials with PCM might be

able to enhance the thermal conductivity of PCMs as for their high surface area to volume ratio [66]. The nanoparticles that can be used with thermal applications are like Aluminum oxide (Al2O3), Copper (Cu), Copper Oxide (CuO), Gold (Au), Silver (Ag), Silicon Carbide (SiC), Titanium Carbide (TiC), Titanium Oxide (TiO2), and CNT [81]. Kaviarasu and Prakash [66] proposed a review on the enhancement in the PCM thermal properties by dispersion of different types of nanoparticles (e.g. Al, Cu, SiO2, CNT, … etc.) referring to these new suspensions as Nano-PCMs. Water could be considered as the highest PCM type in the thermal properties, but has low melting temperature 273.15 K. Regarding the nanoparticles, CNT and CNF showed the highest thermal conductivity. Also, they had found that the volume concentration of these small particles greatly affected the enhancement in thermal conductivity of PCMs, however increasing it more would have bad effect on increasing the viscosity and as a result the pumping power. The storage heat capacity of PCM would be decreased as well. The group claimed that the nanoparticles in PCMs might be more stable than in conventional base fluids.

Sahan and Paksoy [82] carried out an experimental study on the thermo-physical properties of the Nano-PCM for thermal energy storage (TES) applications. They had prepared for the first time the Nano zinc oxide in tubular shape and powder form and suspended it with a paraffin with 10% weight fraction. They had found that the phase-change temperature range of the PCM remained the same while the storage capacity had decreased by around 7%. Hosseini et al. [83] proposed a numerical study on the effect of volume fraction of nanoparticles on the solidification time of Nano-PCM in a vertical enclosure. They had used different materials for nanoparticles like Cu, TiO_2, and Al_2O_3 and they used the water as the PCM material. Applying constant heat flux with insulation from the other sides and changing the value of Grashof number (Gr), they had found that using 0.2 wt% of Cu nanoparticles had decreased the solidification time by around 16% when $Gr = 10^6$ if compared to $Gr = 10^5$. They had concluded that the dispersion of nanoparticles in PCM would enhance the thermal conduction and as a result decrease the solidification time. Sharma and Ganesan [84] had proposed a numerical study on the solidification process of Nano-PCM in a trapezoidal enclosure. Water was used as the phase change material and Cu nanoparticles were used as the dispersed material. Different temperatures were applied on the inclined walls side by side with different volume fractions of the dispersed nanoparticles; 0%, 10%, and 20%. The group used the enthalpy-porosity technique to trace the solid-liquid interface. They had found that increasing the volume fraction would decrease the solidification time, which for example was 3000s for pure water to be 1890s with 10% of Cu nanoparticles and 1300s with 20% loading. In addition, increasing the temperature difference between walls helped in increasing the heat rate especially when dispersing the nanoparticles that helped in enhancing the thermal conduction process.

Induction of such nanomaterials were found as also affecting the rate of heat absorption and rejection. Enhancing the thermal conductivity using the nanoparticles allowed to achieve faster rates of heat absorption and rejection and as a result higher heat transfers to the working fluids. Miqdam et al. [85] proposed a study on the enhancement of PCM thermal conductivity using alumina and titanium oxide nanoparticles. Different weight fractions were applied from 1 wt% up to 5 wt%. Addition of alumina had shown better enhancement in the thermal conductivity of paraffin wax. They had found that the thermal conductivity was inversely proportional with temperature at temperatures lower than the melting point, while it was in direct proportion after exceeding the melting point temperature. Using nanoparticles had shown lower charging and discharging periods as compared to the pure paraffin wax. In addition, increase of weight fraction had an inverse effect on the charging and discharging periods. They have concluded that addition of nanoparticles to PCM had a positive effect on enhancing the thermal conductivity without decreasing the heat capacity. Wu et al. [86] examined addition of different nanoparticles: Cu, Al and C/Cu to melting paraffin for the purpose of heat transfer enhancement. They had noticed that the latent heat of the prepared Cu/paraffin samples was lower than that for pure paraffin and decreased with increase of weight fraction of Cu nanoparticles. In addition, the thermal cycles were found as affecting insignificantly. Furthermore, the heating and cooling periods had been decreased by about 30% and 28% respectively for samples with 1 wt% of Cu. Chaichan et al. [87] investigated an experimental study to evaluate the addition of alumina nanoparticles to paraffin wax. The thermal conductivity was enhanced and that was reflected on the charging and discharging periods. Faster charging and discharging were achieved. In addition, higher temperatures were achieved for the nanoPCM and consequently for the cooling fluid. Furthermore, they noticed that stored energy may be lost faster using nanoparticles, so entering fluid temperature and mass flow rate should be optimized. Al-Waeli et al. [88] studied the combined effect of nanoPCM for storage and nanofluid for cooling on the performance of hybrid PV/T system. They conducted an experimental study to evaluate the effect of different cooling approaches on the performance of the PV/T system. Al-Waeli et al. [88] have applied three different ways for cooling. The first was with a tank filled with water and water flowed through the cooling pipes under the PV panels. In the second way, a tank was filled with PCM and the water flowed through the cooling pipes. In the last way, they filled the tank with SiC /PCM nanoPCM and the cooling pipes carried SiC/water nanofluid. Comparing their performance with a conventional single PV system, using SiC nanoparticles with the PCM and water in the tank and cooling pipes respectively has enhanced the efficiency of the PV system from 8.07% to 13.32%.

Table 6 summarizes the recent studies discussed enhancements in storage and cooling using PCM and NanoPCM.

Finally, Agyenim et al. [89] listed the commercial manufacturer of organic as well as inorganic PCMs as shown in Table 7. From this table, the suitable temperature ranges for low temperature solar collectors could be attained by using PCMs from TEAP, Climator, and Phase Change Products Pty Ltd. (PCP).

To sum up, phase change materials were considered as the best solution for storage of solar energy in the daytime. Benefit of the usage of PCM as an insurance for the thermal stability of the PV cells had been proved from literature. The low thermal conductivity of PCM is the obstacle in the way of using PCM with hybrid PV/T system to assure thermal stability of the PV cells. Integration of nanoparticles with the PCM was considered as a good solution. Few studies had been done on the nanoPCM. Up till now, there are no work has been done on the modality of calculation of the equivalent properties of nanoPCM and the effect of these nanoparticles on the solidification and melting.

9. Life cycle considerations and environmental impact

It is not enough that the PV/T offers high energy output more than that for PV alone, but it should become cost effective by having the additional cost of the thermal part relatively low [90]. However the market of PV/T is still small as compared to the PV and the solar thermal collectors, different commercial designs for PV/T are now available on the market ground [91].

Lamnatou and Chemisana [92] proposed a review on the PV/T investigations especially that related to environmental issues resulted from PV/T technology. Most of the investigations were found to examine the energy payback time EPBT, and CO2 emissions. Many factors affecting the environment were studied like PV cell material, heat transfer fluid, concentrators, alternative materials ... etc. They found that some of the PV cell materials show lower environmental impact during their production, but perform with low efficiency during the system operation. In addition, they clarified that cooling fluid would affect the environmental behavior (e.g. using water in cooling of PV/T instead of air showed better performance due to higher heat extraction). Furthermore, they found that many other parameters affect the environmental performance like alternative PV/T materials, using of

Table 6
Summary of the recent studies related to usage of PCM and NanoPCM in storage and cooling.

Year	Author	PCM or nanoPCM contribution	Results
2017	Su et al. [76]	PCM in cooling of PV/T	- Location of the PCM on the upper side of the cooling fluid duct enhanced the overall performance of the system. - 3 cm thickness is an excellent thickness to get high efficiencies.
2011	Malvi et al. [77]	PCM for storage for PV/T	- Performance enhanced by about 9% when suitable PCM was selected. - Increase in electrical efficiency usually accompanied with decrease in thermal efficiency. - PCM would be better with tuneable melting temperature and high thermal conductivity.
2017	Colla et al. [73]	Alumina and carbon black nanoparticles with paraffin wax PCM (RT20 and RT25)	- Latent heat of PCM enhanced by addition of Al_2O_3 and degraded using CB. - Thermal conductivity of PCM enhanced by addition of CB while it was almost the same using Al_2O_3.
2016	Browne et al. [2]	PCM for storage and water preheating at different arrangements	- Using PCM, heat extraction from PV panel has enhanced by seven times when PCM was used for storage and water preheating.
2011	Huang et al. [78]	PCM in a finned PV panel arrangement for storage and cooling.	- Temperature regulation has enhanced due to natural convection. - The output uniform temperature would positively affect the melting and solidification of PCM.
2010	Hasan et al. [79]	Five different PCM types with different cell sizes PV panels	- Thermal performance depended on thermal mass of PCM and thermal conductivity of PV panel. - PCM type affected the build-up rate of temperature with time. - Highest reduction of PV temperature was achieved with salt hydrate ($CaCl_2$) at different solar radiation levels.
2015	Hasan et al. [3]	Different PCM types to regulate temperature of PV panel	- Higher temperature drop and power saving were achieved using $CaCl_2{\cdot}6H_2O$, 3–4 °C and 3% respectively if compared to capric–palmitic acid in hot climates.
2010	Hasan et al. [80]	Different types of PCM at different thicknesses and solar radiation	- Salt hydrate PCM with 50 mm thickness is the best to keep at least 10 °C of temperature reduction for long period. - Salt hydrate has thermal stability problems for long operations periods.
2015	Atkin and Farid [17]	PCM infused graphite and/or aluminum fins	- PV overall efficiency has increased by 13% using PCM infused graphite with aluminum fins if compared to conventional PV system. - Thermal conductivity of PCM has increased from 0.25 to 16.6 W/m. K by infusing PCM in graphite.
2017	Sahan and Paksoy [82]	Nano zinc oxide nanoparticles in paraffin wax	- Phase change temperature range remained the same. - Heat capacity decreased by 7%.
2014	Hosseini et al. [83]	Cu, TiO_2, or Al_2O_3 dispersed in water as a PCM	- 0.2 wt% of Cu nanoparticles has decreased the solidification time by 16% at Gr = 10^6 if compared to that at Gr = 10^5. - Thermal conduction enhanced and solidification time decreased.
2014	Sharma and Ganesan [84]	Cu nanoparticles dispersed in water as a PCM in a trapezoidal enclosure	- Solidification time decreased as volume fraction increased. - Dispersion of Cu nanoparticles increased conduction and heat rate increased with temperature difference between walls.
2015	Miqdam et al. [85]	Al_2O_3 and TiO_2 nanoparticles dispersed in paraffin wax	- At $T < T_{melting}$: Thermal conductivity was inversely proportional with temperature. - At $T > T_{melting}$: Thermal conductivity was in direct proportion with temperature - Increase of weight fraction had an inverse effect on the charging and discharging periods. - Addition of nanoparticles to PCM enhanced the thermal conductivity without decreasing the heat capacity.
2010	Wu et al. [86]	Cu, Al and C/Cu nanoparticles dispersed in melting paraffin	- Latent heat of the Cu/paraffin samples was lower than that for pure paraffin and decreased with increase of weight fraction of Cu nanoparticles. - Thermal cycles were affecting insignificantly. - Heating and cooling periods had been decreased by about 30% and 28% respectively for samples with 1 wt% of Cu.
2016	Chaichan et al. [87]	Al_2O_3 nanoparticles dispersed in paraffin wax	- Enhancement in thermal conductivity. - Faster charging and discharging were achieved. In addition. - Higher temperatures were achieved for the nanoPCM and outlet working fluid.

Table 7
Commercial manufacturers of PCM [89].

Manufacturer	Temp range (°C)	Types of product	Class
RUBITHERM	− 3–100 °C	Inorganic, organic	Bulk, macro-encapsulated
Cristopia	− 33–27 °C	Inorganic	Micro-encapsulated
TEAP	− 50–78 °C	Inorganic	Bulk, macro-encapsulated
Mitsubishi Chemical	9.5–118 °C	Inorganic	Bulk
Climator	− 18–70 °C	Inorganic	Bulk
EPS Ltd./PCM Products	− 114–164 °C	Inorganic, organic, eutectic	Bulk, macro-/micro-encapsulated
BASF	21–26 °C	Organic	Micro-encapsulated
Phase Change Energy Solutions	23–27 °C	Not available	Macro-encapsulated
Entropy Solutions Inc./PureTemp PCM	− 37–151 °C	Organic	Bulk, macro-/micro-encapsulated
Phase Change Products Pty Ltd. (PCP)	− 21–58 °C	Inorganic	Bulk

reflectors, integration-to-buildings method, applications, recycling, and durability, which were discussed in details in their work paper. Al-Waeli et al. [93] conducted experimental as well as numerical study on a PV/T system using nanofluid while connected to the electricity grid. The study had been done in Malaysia to assess technically and economically the practical application of PV/T systems. They had found that the system capacity factor was (17.82–25.52%). In addition, they calculated the cost of the energy to be about 0.196 USD/kWh with an output electrical efficiency of 9.1%. Furthermore, they had mentioned that such a system would have a payback period of about 7–8 years.

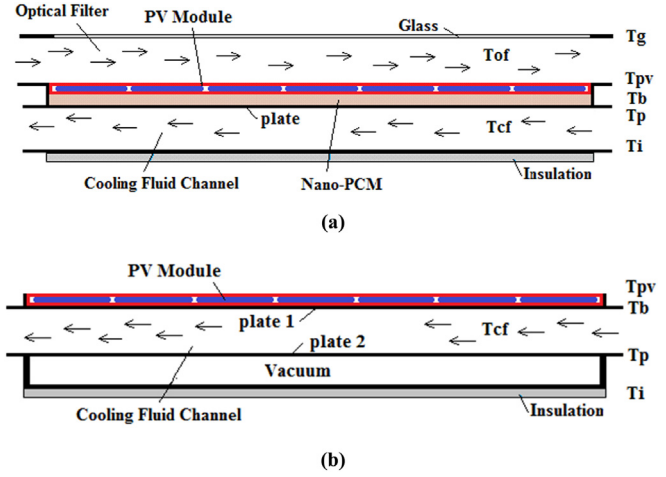


Fig. 32. Schematic diagrams for the case study models.

As enhancement of the performance of PV/T solar systems is still progressing, there is a need for parallel studies that care about the life cycle analysis of the recommended designs and its impact on the environment.

10. Case study

Recently, some researches had been existed to study the effect of integration of both nanofluid and PCM or nanoPCM with PV/T system on the performance of the PV/T system and cooling of the PV panels. Sardarabadi et al. [94] carried out an experimental study to assess the effectiveness of using ZnO/water nanofluid and PCM for cooling of PV panels. Sardarabadi et al. [94]found that the electrical efficiency had increased by about 13% using nanofluid and PCM as compared to conventional PV system. In addition, The average collected thermal energy had been enhanced by about 5% after using ZnO/water nanofluid as compared to deionized water. Furthermore and after exergy analysis, it was found that usage of nanofluid beside the PCM had enhanced the overall exergy efficiency by about 23% if compared to conventional PV system. Al-Waeli et al. [95] investigated an experimental study on the integration of PCM and nanofluid with hybrid PV/T systems. The nanofluid was circulating in tubes through the PCM located beneath the PV panel. At flow rate of 0.17 kg/s, Al-Waeli et al. [95] have found that the electrical efficiency had enhanced from 7.1% to 13.7%. In addition, the utilization of the thermal energy had scored 72%. Al-Waeli et al. [96] had reviewed the work done during the last

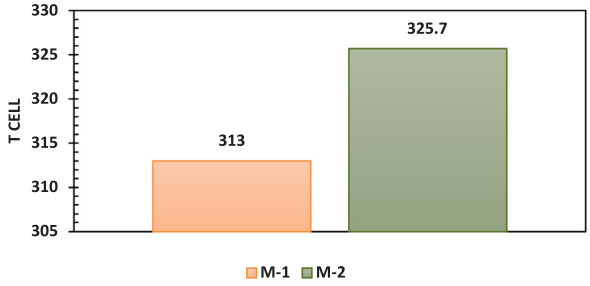


Fig. 33. PV cell temperature for the case study models.

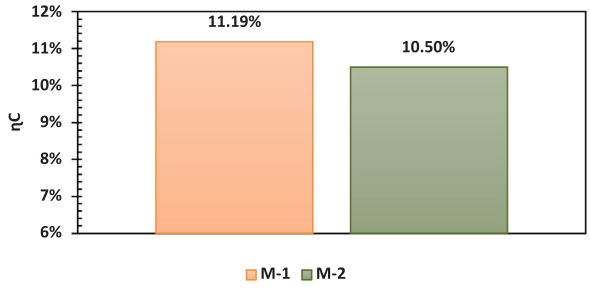


Fig. 34. Electric conversion efficiency for the case study models.

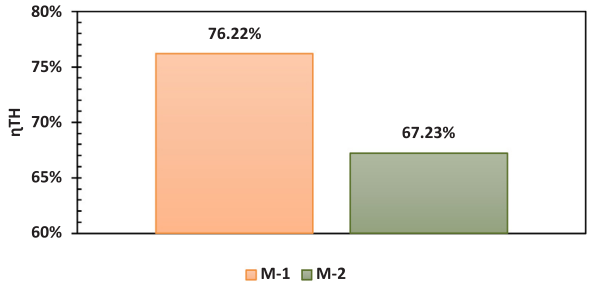


Fig. 35. Thermal efficiency for the case study models.

Table 8
Governing equations of the case study models.

M- #	Balance	Governing Equations	Equation no.
1	Glass cover	$\alpha_g G = h_{g-am}(T_g - T_{am}) + h_{g-of}(T_g - \bar{T}_{of})$	Eq. 1
	Optical fluid	$\tau_g\alpha_{of} G = h_{of-g}(\bar{T}_{of} - T_g) + h_{of-pv}(\bar{T}_{of} - T_{pv}) + \frac{m^o_{of} c_{pof}}{W}\frac{dT_{of}}{dx_1}$	Eq. 2
	PV panel	$\alpha_{pv}\tau_g\tau_{of} G = h_{pv-of}(T_{pv} - \bar{T}_{of}) + \frac{T_{pv} - T_b}{R_b} + \eta_c\alpha_{pv}\tau_g\tau_{of} G$	Eq. 3
	Back plate	$\frac{T_{pv} - T_b}{R_b} = k_{PCM}\frac{T_b - T_{p1}}{\Delta t_{PCM}}$	Eq. 4
	Plate 1	$k_{PCM}\frac{T_b - T_p}{\Delta t_{PCM}} = h_{p-cf}(T_p - \bar{T}_{cf})$	Eq. 5
	Cooling fluid	$h_{p-cf}(T_p - \bar{T}_{cf}) = h_{cf-I}(\bar{T}_{cf} - T_I) + \frac{m^o_{cf} c_{pcf}}{L}\frac{dT_{cf}}{dx_2}$	Eq. 6
	Insulation	$h_{cf-I}(\bar{T}_{cf} - T_I) = h_{I-am}(T_I - T_{am})$	Eq. 7
	Efficiency	$\eta_c = \eta_{ref}[1-\beta_{ref}(T_c - T_{ref})]$	Eq. 8
2	PV panel	$\alpha_g G = h_{pv-am}(T_{pv} - T_{am}) + \frac{T_{pv} - T_b}{R_b} + \eta_c\alpha_g G$	Eq. 9
	Back plate	$\frac{T_{pv} - T_b}{R_b} = h_{b-cf}(T_b - \bar{T}_{cf})$	Eq. 10
	Cooling fluid	$h_{b-cf}(T_b - \bar{T}_{cf}) = h_{cf-p}(\bar{T}_{cf} - T_p) + \frac{m^o_{cf} c_{pcf}}{L}\frac{dT_{cf}}{dx_2}$	Eq. 11
	Plate	$h_{cf-p}(\bar{T}_{cf} - T_p) = h_{r,p-I}(T_p - T_I)$	Eq. 12
	Insulation	$h_{r,p-I}(T_p - T_I) = h_{I-am}(T_I - T_{am})$	Eq. 13
	Efficiency	$\eta_c = \eta_{ref}[1-\beta_{ref}(T_c - T_{ref})]$	Eq. 14

four decades for improvement of PV/T system and paid more attention on how the PV/T overall performance had been improved as a result of enhancing the thermal performance. They had found that the recent studies in the past seven years was concentrating on the improvement of PV/T performance using novel techniques for integration of nanofluid and PCM, which still have a lot of work to be done. Sathe and Dhoble [97] as well proposed a review on the developments in the hybrid PV/T topic during the last seven years. They covered the numerical and experimental work done on integration of nanofluid and PCM in PV/T systems. They had mentioned that nanofluid based PV/T systems had very limited experimental studies. In addition, they recommended that using nanofluid for both cooling and filtration would propose considerable enhancement in PV/T performance. Furthermore, they had found that PCM was very effective in attaining uniform temperature distribution throughout the PV.

In this section, a simple analytical comparison between a hybrid PV/T system comprising optical nanofluid, cooling nanofluid, and nanoPCM and a traditional PV/T system using nanofluid for cooling will be presented to show the expected enhancement in the performance. The governing energy equations were developed and solved analytically using EES program.

10.1. Physical models

The modified model (M-1) consists of a PV module, optical fluid transparent channel above the PV, nanoPCM layer directly beneath the PV panel and cooling fluid channel below of the nanoPCM layer in addition to the insulation on the bottom side. The model used water/silver as an optical fluid, paraffin wax/graphene as nanoPCM and water/graphene as a cooling fluid. In the traditional model (M-2), only water/graphene channel existed below the PV module. An air gap was added in M-2 below the cooling channel to provide more insulation as a challenge to prove the advance of M-1.

10.2. Working principle

Fig. 32-(a) shows the configuration of M-1. The incident solar radiation falls on the glass cover, the useful radiation to the PV will pass through the optical fluid to reach the PV cells and the rest should be absorbed in the optical fluid. As temperature of the PV panel increases, part of the heat should be stored in the nanoPCM layer and the rest will be carried out by the cooling fluid as well as the optical fluid. In M-2, heat of the PV panel is carried out using the cooling water/graphene nanofluid as shown in Fig. 32-(b).

10.3. Mathematical model

The performance of the PV/T models has been evaluated through the calculation of the different temperatures, thermal, and electrical efficiencies using previously calculated optical and thermal properties. The governing equations of the two models are shown in Table 8.

10.4. Results

The PV cell temperature, thermal and electrical conversion efficiencies have been used as points of comparison between the two models. It can be seen clearly from Fig. 33 that high temperature reduction by about 13 degree can be attained using M-1. Fig. 34 shows that the electrical conversion efficiency has been enhanced from 10.5% to 11.9% using M-1. Finally, larger amount of heat collection can be noticed from Fig. 35 as a result of using M-1. The thermal efficiency has increased by around 9%.

11. Conclusions

In the present paper, a review of the recent studies that covers the design and cooling technologies for the hybrid PV/T system has been provided. A case study comparing between the proposed modifications and a traditional PV/T system has been provided at the previous section. Different related issues have been discussed leading to the following conclusions:

- ✓ Hybrid PV/T system is considered as an excellent solution get the best use of solar energy for electrical and thermal energy production.
- ✓ The high temperature of the PV panel is an important problem that should be tackled seriously.
- ✓ Cooling from both the top and bottom sides of the PV is the best method to collect the largest heat amount.
- ✓ Using liquid channels, other than circular ones like honeycomb channels; enhanced the cooling rate and provided uniform temperature distribution throughout the fluid as well.
- ✓ Using nanofluid in cooling has showed an efficient alternative for the traditional liquids like water and EG due to its modified thermal properties.
- ✓ Using special nanofluid in optical filtration is a new promising technique that has proven high capability to control the solar radiation transmitted to the PV and that absorbed in the fluid itself.
- ✓ Usage PCM below the PV panels is very important as it provides storage of the thermal energy and uniform temperature for the panels.
- ✓ Dispersion of nanoparticles in PCM (NanoPCM) is considered as a new promising solution to overcome the low thermal conductivity problem of the PCM.

To sum up, we can claim that usage of hybrid PV/T system with optimized application of nanofluids ,in cooling and optical filtration; and integration of nanoPCM ,for storage and cooling; with modified channel designs can be considered as the best solution at the moment to make the best use of solar energy.

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